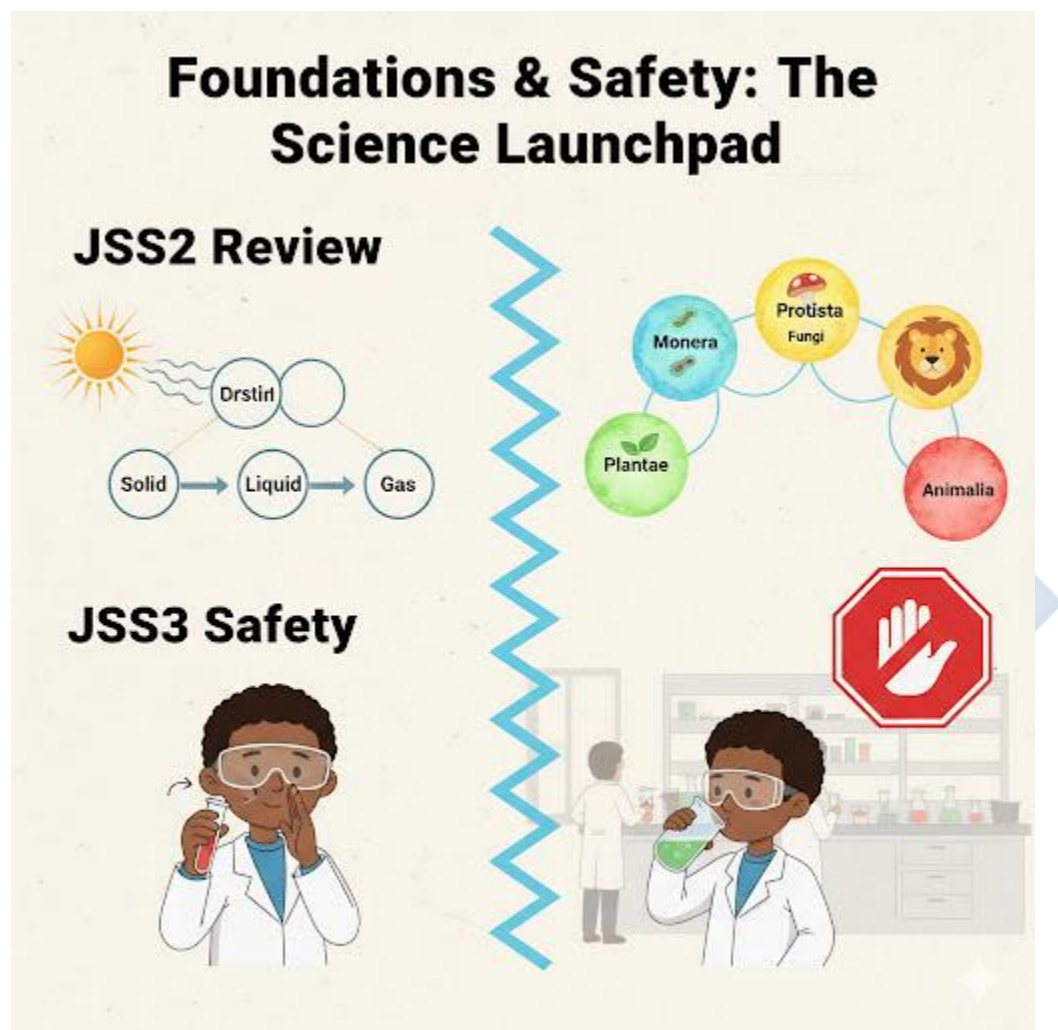


JSS 3 INTERMEDIATE SCIENCE LESSON NOTES

FIRST TERM

WEEK 1: REVIEW OF JSS 2 WORK



Content: (I) Revision of matter, energy, and classification (II) Laboratory conduct and safety (III) Introduction to JSS 3 concepts.

Performance Objectives: By the end of this lesson, students should be able to:

1. Recall the definitions of matter, energy, and their forms.
2. Summarize the five-kingdom classification of living things.

3. List at least ten (10) laboratory safety rules.
4. Differentiate between the key JSS2 topics and the upcoming JSS3 topics.

Lesson Content:

1. Review of Key JSS2 Concepts

- **A. Matter:**

- **Definition:** Matter is anything that has **mass** and occupies **space (volume)**.
- **States:** Solid (definite shape/volume), Liquid (indefinite shape, definite volume), Gas (indefinite shape/volume).
- **Classification:**
 - **Elements:** Pure substances that cannot be broken down (e.g., Iron, Oxygen, Carbon).
 - **Compounds:** Two or more elements chemically combined (e.g., Water - H_2O , Carbon Dioxide - CO_2).
 - **Mixtures:** Two or more substances physically combined, not chemically (e.g., sand and water, salt solution).
- **Separation Techniques:** We learned about filtration, evaporation, sieving, and decantation.

- **B. Energy:**

- **Definition:** Energy is the **ability or capacity to do work**.
- **Forms (Recap):** Potential (stored), Kinetic (movement), Heat, Light, Sound, Electrical, Chemical.
- **Principle of Conservation:** Energy cannot be created or destroyed, but can be transformed from one form to another (e.g., a battery (chemical) powers a bulb (electrical) which gives off light and heat).

- **C. Classification of Living Things (The Five Kingdoms):**

1. **Monera:** Unicellular, no true nucleus (prokaryotic). (e.g., Bacteria, Blue-green algae).
2. **Protista:** Unicellular, have a nucleus (eukaryotic). (e.g., Amoeba, Paramecium).

3. **Fungi:** Multicellular, "plant-like" but cannot make their own food (heterotrophic). (e.g., Mushroom, Mould, Yeast).
4. **Plantae:** Multicellular, make their own food (autotrophic) through photosynthesis. (e.g., Mango tree, maize, fern).
5. **Animalia:** Multicellular, cannot make their own food (heterotrophic), and can move. (e.g., Insects, fish, lions, humans).

2. Laboratory Conduct and Safety (Recap) The laboratory is a place for scientific discovery, but it can be dangerous. Safety is the *most* important rule.

- **Rules:**

1. **NEVER** enter the lab without a teacher or lab attendant.
2. **NO** eating, drinking, or playing in the lab.
3. Wear protective clothing (lab coat, safety goggles) when required.
4. Read all instructions *before* starting an experiment.
5. **NEVER** taste, smell, or touch chemicals unless instructed.
6. When smelling a chemical, *waft* the vapour towards your nose; do not sniff directly from the container.
7. Report all accidents, breakages, or spills to the teacher *immediately*.
8. Always point test tubes *away* from yourself and others when heating.
9. Clean your workspace and wash your hands before leaving the lab.
10. Know the location of safety equipment (fire extinguisher, first aid box, eye-wash station).

3. Introduction to JSS 3 Concepts This term (First Term), we will focus on our planet and how we live on it. We will move from basic classifications to understanding large-scale systems.

- **Ecosystems:** How living things (like us) interact with non-living things (like water and air).
- **Natural Resources:** What our country is blessed with (oil, minerals, forests) and how we must protect them.

- **Environmental Hazards:** Problems like flooding, erosion, and drought, and how we can prevent them.
- **Sustainable Development:** How we can live well today *without* destroying the planet for future generations.

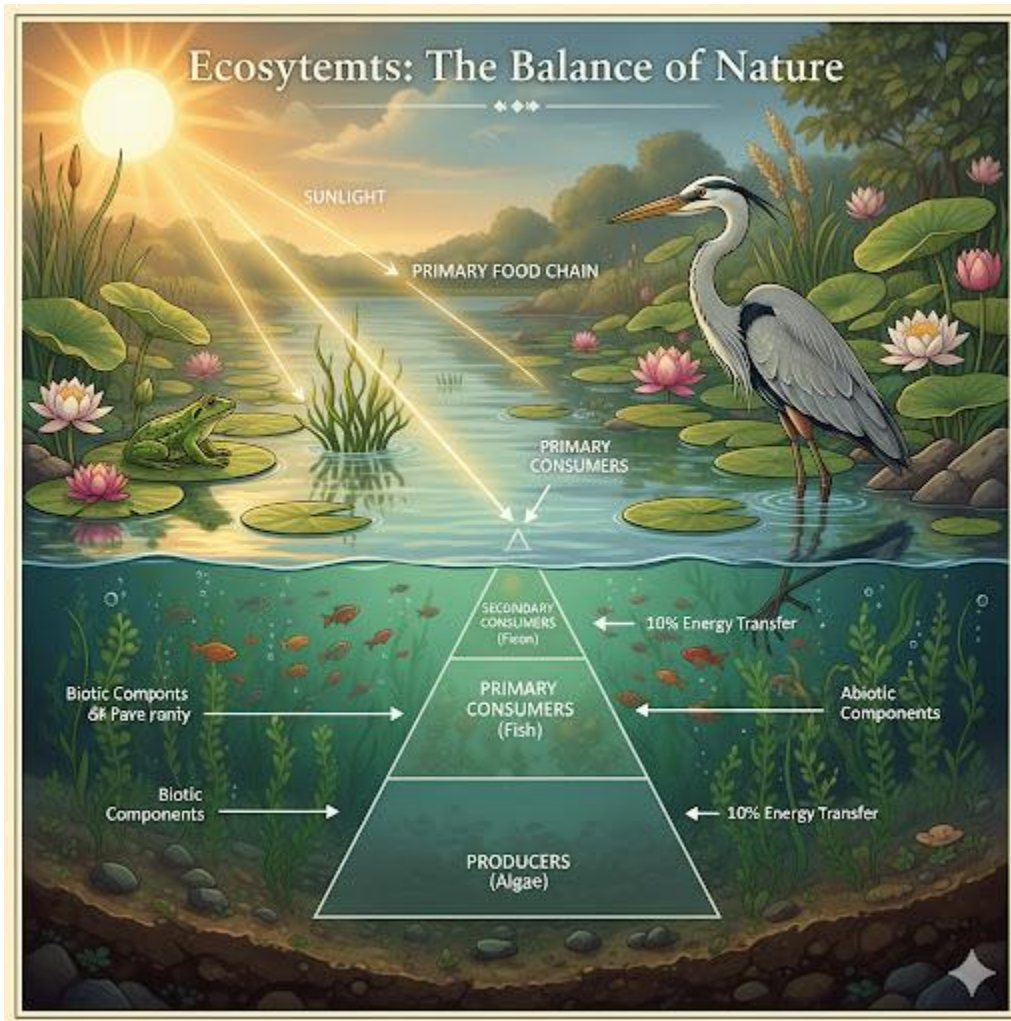
Evaluation:

1. What is the main difference between a compound and a mixture?
2. State the Principle of Conservation of Energy.
3. Name the five (5) kingdoms of living things.
4. Why is it forbidden to eat or drink in the laboratory?
5. What is the *first* thing you should do if you break a beaker in the lab?

Assignment:

1. Define Matter and Energy.
2. List the five kingdoms of living things and give one example for each.
3. Write down five (5) laboratory safety rules you think are the most important.
4. Briefly explain the difference between a renewable resource (like sunlight) and a non-renewable resource (like crude oil).

WEEK 2: ECOSYSTEM



Content: (I) Meaning and components (II) Interdependence and balance of nature (III) Food chains, food webs, and energy flow (IV) Disruption of ecosystems.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Ecosystem" and list its two main components (biotic and abiotic).
2. Provide at least three examples of biotic and three examples of abiotic components.
3. Draw a simple food chain and a food web, labeling the producers, consumers, and decomposers.
4. Explain the "balance of nature" and how energy flows from the sun to all living things.

5. List two human activities that can disrupt an ecosystem.

Lesson Content:

1. Meaning and Components of an Ecosystem

- **Definition:** An **Ecosystem** is a biological community of living organisms (biotic) interacting with their non-living physical environment (abiotic) in a particular area. An ecosystem can be as large as a desert or an ocean, or as small as a pond or a rotting log.
- **Components:**
 - **A. Biotic (Living) Components:** All the living or once-living parts of the ecosystem. They are grouped by how they get energy:
 1. **Producers (Autotrophs):** Make their own food, usually from sunlight. (e.g., **Plants**, algae).
 2. **Consumers (Heterotrophs):** Get energy by eating other organisms.
 - **Primary Consumers (Herbivores):** Eat producers. (e.g., Grasshopper, cow, rabbit).
 - **Secondary Consumers (Carnivores/Omnivores):** Eat primary consumers. (e.g., Frog, fox).
 - **Tertiary Consumers:** Eat secondary consumers. (e.g., Snake, hawk).
 3. **Decomposers (Saprotrophs):** Break down dead organisms and waste products, returning nutrients to the soil. (e.g., **Bacteria, Fungi**).
 - **B. Abiotic (Non-Living) Components:** All the non-living chemical and physical parts of the environment. (e.g., **Sunlight, Water, Air, Soil, Temperature, Salinity**).

2. Interdependence and Balance of Nature

- **Interdependence:** All these components depend on each other.
 - Plants (biotic) need soil, water, and sunlight (abiotic) to grow.
 - Animals (biotic) need plants (biotic) for food and water (abiotic) to drink.
 - Decomposers (biotic) need dead animals and plants (biotic) to feed on.

- Plants (biotic) need decomposers (biotic) to create nutrients in the soil (abiotic).
- **Balance of Nature:** In a healthy ecosystem, the population of each organism is kept in check. There is a balance between producers, consumers, and decomposers. If one part is removed, the whole system can be affected. (e.g., If all the frogs are removed, the grasshopper population might explode, destroying the grass).

3. Food Chains, Food Webs, and Energy Flow

- **Energy Flow:** The *ultimate* source of all energy in (almost) every ecosystem is the **SUN**. Energy flows in *one direction*.
- **Food Chain:** A simple, single pathway of energy flow in an ecosystem.
 - *Example:* Grass → Grasshopper → Frog → Snake → Hawk (Producer) → (Primary) → (Secondary) → (Tertiary) → (Quaternary)
- **Food Web:** A more complex and realistic model that shows all the *interconnected* food chains in an ecosystem. A food web shows that one animal (like a hawk) might eat snakes, rats, *and* frogs.



- **Energy Transfer:** When a grasshopper eats grass, it only gets about 10% of the grass's energy. The other 90% is lost as heat or used for living. This means the Hawk at the top gets very little of the original sun's energy.

4. Disruption of Ecosystems When the "balance of nature" is disturbed, the ecosystem can be damaged. This is often caused by human activities:

1. **Deforestation:** Cutting down trees removes producers, destroys habitats, and leads to soil erosion.
2. **Pollution:** (To be covered in Week 4) Chemicals and waste kill plants and animals.
3. **Overhunting/Overfishing:** Removing too many consumers from a food web, causing an imbalance.
4. **Introducing Invasive Species:** Adding a new animal or plant that has no natural predators can cause it to take over.

Evaluation:

1. What is an ecosystem?
2. What are the two main components of an ecosystem?
3. List three biotic and three abiotic components of a school garden.
4. Draw a simple food chain with 4 levels.
5. What is the ultimate source of energy for most ecosystems?

Assignment:

1. Define Producer, Consumer, and Decomposer.
2. Draw a simple **food web** (not a chain) for a local environment (e.g., a field or pond). It must have at least 5 organisms.
3. What is the "balance of nature"?
4. What is the "10% rule" of energy transfer?
5. Describe two human activities that can disrupt an ecosystem.

WEEK 3: NATURAL RESOURCES



Content: (I) Types: renewable and non-renewable (II) Importance to national economy (III) Conservation and management (IV) Field observation project.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Natural Resources."
2. Differentiate between renewable and non-renewable resources, giving two examples of each.
3. Explain the importance of natural resources (like crude oil, limestone, and forests) to Nigeria's economy.

4. Define "Conservation" and list three methods of conserving resources.
5. Conduct a simple field observation to identify local resources.

Lesson Content:

1. What are Natural Resources?

- **Definition:** Natural Resources are materials or substances that exist **naturally** in the environment and are considered valuable to humans. They are "gifts of nature" that we use for survival and economic development.
- *Examples in Nigeria:* Crude oil, natural gas, coal, tin, limestone, forests, water, sunlight, fertile soil.

2. Types of Natural Resources

- **A. Renewable Resources:**
 - **Definition:** Resources that can be replaced or replenished naturally over a *short* period of time. They will not run out if we manage them wisely.
 - **Examples:**
 - **Sunlight (Solar Energy):** Will not run out.
 - **Water (Hydropower):** Replenished by the water cycle.
 - **Wind:** Will not run out.
 - **Forests/Trees:** Can be replanted (this is "managed" renewable).
 - **Living Organisms:** (Crops, Animals) Can be regrown/reproduced.
- **B. Non-Renewable Resources:**
 - **Definition:** Resources that exist in a **fixed amount** and are formed over *millions of years*. Once they are used up, they are gone forever (or for a very, very long time).
 - **Examples:**
 - **Fossil Fuels:** (Formed from dead plants/animals)
 - **Crude Oil (Petroleum):** The backbone of Nigeria's economy.
 - **Natural Gas:** Often found with oil.
 - **Coal:** Found in places like Enugu.

- **Minerals:** (e.g., **Tin** (Jos), **Limestone** (for cement), **Iron Ore** (Ajaokuta), **Gold**).

3. Importance to National Economy

- **1. Government Revenue:** Nigeria's government gets most of its money by selling crude oil. This money is used to build roads, schools, and hospitals.
- **2. Foreign Exchange:** When Nigeria sells oil and gas to other countries, it earns foreign currency (like US Dollars), which it uses to buy things it doesn't produce.
- **3. Raw Materials:** Resources are the raw materials for industries.
 - Limestone → Cement Industry
 - Forests (Timber) → Furniture and Paper Industry
 - Crude Oil → Petrol, Kerosene, Plastics
- **4. Employment:** The oil and gas, mining, and agriculture sectors create millions of jobs.

4. Conservation and Management

- **Definition: Conservation** is the wise, careful, and efficient use and protection of natural resources to prevent waste, loss, or depletion (running out).
- **Why Conserve?** To ensure that these resources are available for *future generations* (this is **Sustainability**).
- **Methods of Conservation:**
 - 1. Reduce, Reuse, Recycle (The 3 R's):**
 - **Reduce:** Use less (e.g., turn off lights, use less water).
 - **Reuse:** Use items multiple times (e.g., reuse plastic bags, water bottles).
 - **Recycle:** Turn waste materials into new products (e.g., recycling plastic, glass, paper).
 - 2. Afforestation:** Planting new trees to replace those that were cut down.
 - 3. Using Alternatives:** Shifting from non-renewable (like petrol) to renewable (like solar power) for energy.
 - 4. Government Laws:** Creating national parks to protect forests and wildlife; punishing illegal mining or oil spillage.

5. Field Observation Project (Activity)

- Students will walk around the school compound or their local area for 15 minutes.
- They must list all the natural resources they can identify.
- *Examples:*
 - Sunlight (Renewable)
 - Soil (Renewable, if managed)
 - Trees/Plants (Renewable)
 - Water (from a tap/well) (Renewable)
 - Wind (Renewable)
 - Stones/Gravel (Non-Renewable)

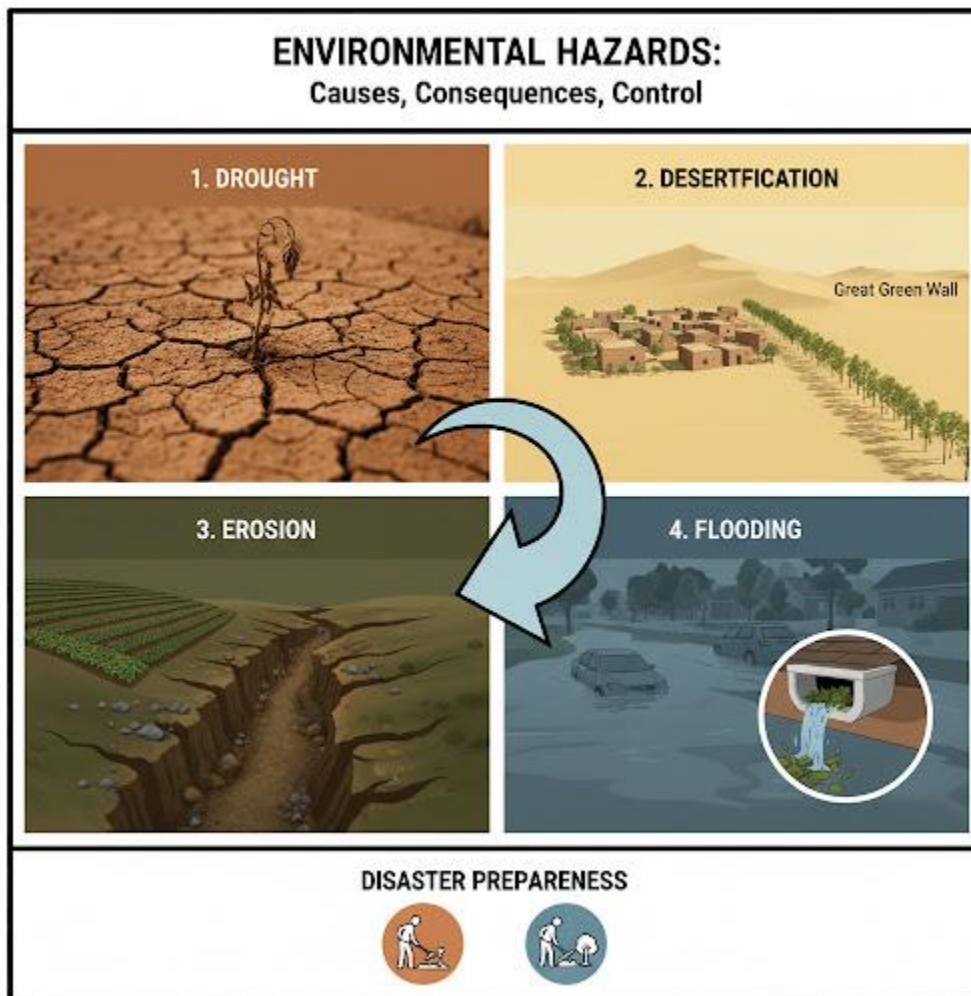
Evaluation:

1. What are natural resources?
2. What is the main difference between a renewable and a non-renewable resource?
3. Is crude oil renewable or non-renewable? Why?
4. Explain two ways natural resources are important to Nigeria's economy.
5. What is "conservation"?

Assignment:

1. List three (3) renewable resources and three (3) non-renewable resources found in Nigeria.
2. Define "Afforestation."
3. Explain the 3 R's (Reduce, Reuse, Recycle) with one example for each.
4. Why is it dangerous for a country (like Nigeria) to depend on only *one* natural resource (like crude oil)?

WEEK 4: ENVIRONMENTAL HAZARDS



Content: (I) Meaning and types: drought, desertification, erosion, flooding (II) Causes and prevention (III) Disaster preparedness.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "environmental hazard."
2. Identify and describe the four major hazards: drought, desertification, erosion, and flooding.
3. State at least one major cause and one prevention method for each hazard.
4. Define "disaster preparedness."

Lesson Content:

1. Meaning of Environmental Hazard

- An **Environmental Hazard** is a natural or man-made event that has the potential to cause harm to people, property, and the environment.
- When a hazard *actually* causes harm, it is called a **disaster**.

2. Types of Environmental Hazards in Nigeria

- **A. Drought:**
 - **Meaning:** A long period of time with abnormally low rainfall, leading to a severe shortage of water.
 - **Causes:** Natural lack of rain (climate patterns); human activities like deforestation (trees help create rain).
 - **Effects:** Crop failure, famine (lack of food), death of livestock, drying up of rivers (like Lake Chad).
 - **Prevention/Control:**
 1. **Irrigation:** Building systems to bring water to farms.
 2. **Afforestation:** Planting trees (e.g., the "Great Green Wall" project) to attract rain and stop desert spread.
 3. **Dams:** Building dams to store water during rainy seasons.
- **B. Desertification:**
 - **Meaning:** The process by which fertile land (good for farming) in dry regions turns into a desert. This is a major problem in Northern Nigeria (Sokoto, Kano, Borno, etc.).
 - **Causes:**
 1. **Drought** (see above).
 2. **Overgrazing:** Allowing too many animals (cows, goats) to eat all the grass, leaving the soil bare.
 3. **Deforestation:** Cutting down trees for firewood, which exposes the soil to the sun and wind.

- **Prevention/Control:**
 1. **Afforestation:** Planting trees (especially "shelter belts") to block the wind.
 2. **Controlled Grazing:** Moving animals from one plot to another to allow grass to regrow.
- **C. Erosion (Soil Erosion):**
 - **Meaning:** The washing or blowing away of the topsoil (the most fertile part of the soil).
 - **Types:**
 - **Wind Erosion:** Wind blows dry, bare soil away (common in the North).
 - **Water Erosion:** Rain washes soil away. This is a *massive* problem in Southeastern Nigeria (e.g., the Agulu-Nanka gully erosion in Anambra).
 - **Causes:** Deforestation, overgrazing, poor farming methods (like tilling down a slope).
 - **Prevention/Control:**
 0. **Cover Cropping/Planting Trees:** Plant roots hold the soil together.
 1. **Terracing:** Cutting "steps" into a hillside to slow down water.
 2. **Contour Ploughing:** Ploughing *across* a slope, not down it.
- **D. Flooding:**
 - **Meaning:** An overflow of a large amount of water onto land that is normally dry. This is a major problem in coastal cities (like Lagos) and along rivers (like Benue).
 - **Causes:**
 0. **Heavy Rainfall.**
 1. **Poor Drainage:** Blocked gutters and canals (often by human waste).
 2. **Building on Waterways:** People building houses on land meant to be a drainage channel.
 - **Prevention/Control:**
 0. **Good Drainage:** Constructing and *clearing* gutters and canals.
 1. **Legislation:** Banning construction on waterways.
 2. **Dams:** To control river water flow.

3. Disaster Preparedness

- **Definition:** Taking steps *before* a hazard strikes to reduce its impact.
- **Examples:**
 - Listening to warnings (e.g., from NIMET - Nigerian Meteorological Agency).
 - Educating the public on what to do (e.g., move to higher ground during a flood).
 - Storing emergency food and water.
 - Clearing your gutters before the rainy season.

Evaluation:

1. What is an environmental hazard?
2. What is the difference between drought and desertification?
3. What is soil erosion?
4. List two human causes of flooding in Lagos.
5. What is "disaster preparedness"?

Assignment:

1. Create a table with four columns: Hazard (Drought, Desertification, Erosion, Flooding), Meaning, Cause, and Prevention.
2. Fill in the table with at least one point for each box.
3. What is the "Great Green Wall" project? (A quick research question).
4. What is the main type of environmental hazard in your specific local area? (Flooding, erosion, etc.).

WEEK 6: ENVIRONMENTAL HEALTH



Content: (I) Meaning and components (II) Importance of a clean environment (III) Role of individuals and government (IV) Environmental health officers.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Environmental Health."
2. List the key components of a healthy environment.
3. State three reasons why a clean environment is important for human health.
4. Describe the role of individuals and the government in maintaining environmental health.
5. Identify the role of an Environmental Health Officer ("Wole-Wole").

Lesson Content:

1. Meaning and Components

- **Definition: Environmental Health** is the branch of public health that focuses on the relationships between people and their environment. It aims to promote human health and well-being by ensuring the environment is safe and clean.
- **Components of a Healthy Environment:**
 1. **Clean Air:** Free from smoke, dust, and chemical pollutants.
 2. **Safe and Clean Water:** Free from germs, chemicals, and waste (potable water).
 3. **Effective Waste Management:** Proper collection and disposal of refuse and sewage.
 4. **Safe Food:** Food that is free from contamination (germs, chemicals).
 5. **Vector Control:** Controlling pests that spread disease (e.g., mosquitoes, rats, flies).
 6. **Noise Control:** An environment free from harmful or stressful levels of noise.

2. Importance of a Clean Environment A clean environment is not just "nice to have"; it is *essential* for human life.

1. **Prevents Disease:** This is the most important.
 - Clean water prevents **cholera** and **typhoid**.
 - Vector control prevents **malaria** (mosquitoes) and **Lassa fever** (rats).
 - Clean air reduces respiratory problems like **asthma**.
2. **Improves Quality of Life:** A clean, quiet, and green environment reduces stress and improves mental health.
3. **Economic Benefits:** People are healthier, so they are more productive at work and school. The government spends less money on treating preventable diseases.
4. **Protects the Ecosystem:** A clean environment supports local wildlife (birds, fish, etc.).

3. Roles in Maintaining Environmental Health

- **A. Role of the Individual (Your Role):**

1. **Proper Waste Disposal:** Do NOT throw trash in the gutter. Bag your waste properly.
 2. **Personal Hygiene:** Keep your body, clothes, and home clean.
 3. **Sanitation:** Clear your gutters; don't allow stagnant water (where mosquitoes breed).
 4. **Reduce Pollution:** Avoid burning trash; walk or use public transport instead of cars for short trips.
 5. **Report Issues:** Report broken sewage pipes or large waste dumps to the authorities.
- **B. Role of the Government (Local, State, Federal):**
 1. **Provide Infrastructure:** Build water treatment plants, drainage systems, and landfills.
 2. **Waste Management:** Organize and fund agencies (like LAWMA in Lagos) to collect and dispose of waste.
 3. **Create and Enforce Laws:** Set pollution limits for factories; arrest people for improper waste disposal.
 4. **Public Education:** Run campaigns to teach people the importance of hygiene.
 5. **Employ Health Officers:** To inspect the community and enforce standards.

4. Environmental Health Officers

- These are the (formerly called "Sanitary Inspectors" or "Wole-Wole" in Yoruba).
- **Role:** They are government workers who:
 - Inspect houses, markets, and restaurants to ensure they are clean.
 - Enforce sanitation laws (e.g., during "Environmental Sanitation Day").
 - Educate the public on hygiene.
 - Control pests and disease outbreaks.

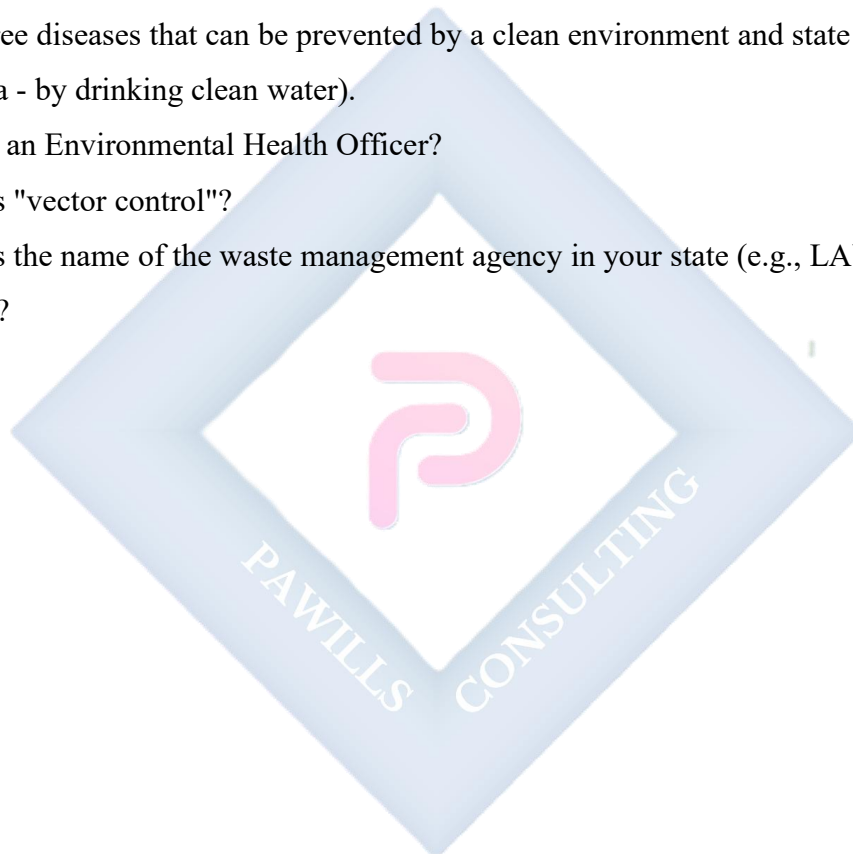
Evaluation:

1. What is environmental health?
2. List four components of a healthy environment.

3. What is the *most* important reason for maintaining a clean environment? (Prevents disease).
4. List two roles of an individual in environmental health.
5. List two roles of the government in environmental health.

Assignment:

1. List three diseases that can be prevented by a clean environment and state how. (e.g., Cholera - by drinking clean water).
2. Who is an Environmental Health Officer?
3. What is "vector control"?
4. What is the name of the waste management agency in your state (e.g., LAWMA in Lagos)?



WEEK 8: WASTE DISPOSAL AND RECYCLING



Content: (I) Methods of waste disposal (II) Importance of recycling (III) Practical waste sorting (IV) Effects of poor waste management.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "waste" and list the main types (solid, liquid, gaseous).
2. Describe at least three modern methods of waste disposal.
3. Define "Recycling" and state its importance.
4. Explain the "3 R's" (Reduce, Reuse, Recycle).
5. List three effects of poor waste management.

Lesson Content:

1. Meaning and Types of Waste

- **Waste (Refuse):** Any material that is no longer wanted or needed and is thrown away.
- **Types of Waste:**
 1. **Solid Waste (Refuse):** Garbage from homes, schools, and markets (e.g., plastic bottles, food scraps, paper, cans).
 2. **Liquid Waste (Sewage/Effluent):** Wastewater from sinks, toilets, and bathrooms (sewage); and chemical waste from factories (effluent).
 3. **Gaseous Waste:** Harmful gases released from factories, car exhaust, and burning trash (e.g., carbon monoxide).

2. Methods of Waste Disposal

- **A. Bad/Unsafe Methods:**
 - **Dumping in Gutters/Rivers:** Causes flooding and water pollution.
 - **Uncontrolled Burning:** Causes air pollution.
- **B. Proper/Modern Methods:**
 1. **Landfill (Sanitary Landfill):** The *most common* modern method. A large, deep pit is dug *far* from the city. Waste is dumped in thin layers and covered with soil each day. This prevents smells and stops rats/flyes from breeding.
 2. **Incineration:** Burning waste at *very high temperatures* (over 1000°C) in a controlled factory. This reduces the volume of waste by 90% and can be used to generate electricity (called "Waste-to-Energy").
 3. **Recycling:** (See below).
 4. **Composting:** (For organic waste) Allowing biodegradable waste (food scraps, leaves) to rot down and be used as fertilizer.

3. The 3 R's: The Waste Hierarchy This is the *best* strategy for managing waste, in order of importance.

1. **REDUCE:** The best solution. Use *less* in the first place.
 - *Example:* Buy products with less packaging. Don't take a plastic bag if you can carry the item.
2. **REUSE:** Use an item more than once.
 - *Example:* Reuse glass jars for storage. Refill your water bottle. Give old clothes to someone else.
3. **RECYCLE:** The last option.
 - **Definition:** The process of converting waste materials into new materials and objects.
 - **Importance:**
 - Conserves natural resources (e.g., recycling paper saves trees).
 - Saves energy (it takes less energy to recycle aluminum than to make it from raw ore).
 - Reduces the amount of waste in landfills.
 - **Common Recyclables:** Plastic (PET bottles), glass, paper, aluminum cans, steel.

4. Effects of Poor Waste Management (This is linked to Week 7 - Environmental Health)

1. **Disease Outbreaks:** Piles of waste attract rats, flies, and mosquitoes, which spread diseases like Lassa fever, cholera, and malaria.
2. **Flooding:** Waste blocks gutters and drains, causing severe flooding.
3. **Pollution:** Waste pollutes water (making it unsafe to drink) and air (from burning).
4. **Ugly Environment:** It makes the community look dirty and unpleasant, which can reduce property values and tourism.

5. Practical Waste Sorting Activity (Teacher brings in a bag of clean, mixed waste: e.g., plastic bottle, paper, old cloth, food wrapper, banana peel).

- Students will sort the items into bins:
 - **Recyclable:** (Plastic bottle, paper)
 - **Reusable:** (Old cloth - as a rag)
 - **Compostable (Organic):** (Banana peel)

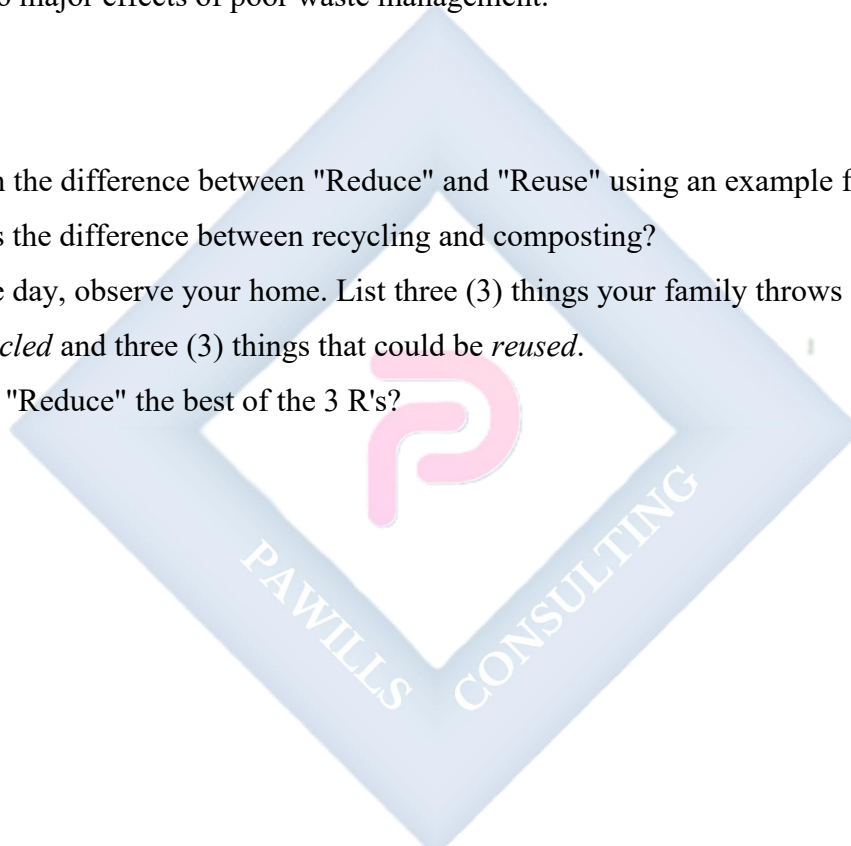
- **Landfill (General Waste):** (Food wrapper)

Evaluation:

1. Define waste.
2. List the 3 R's in their order of importance.
3. What is a sanitary landfill?
4. What is recycling?
5. List two major effects of poor waste management.

Assignment:

1. Explain the difference between "Reduce" and "Reuse" using an example for each.
2. What is the difference between recycling and composting?
3. For one day, observe your home. List three (3) things your family throws away that could be *recycled* and three (3) things that could be *reused*.
4. Why is "Reduce" the best of the 3 R's?



WEEK 9: CLIMATE AND ENVIRONMENTAL CHANGES



Content: (I) Meaning and causes of climate change (II) Effects of global warming (III) Mitigation and adaptation (IV) School-based tree planting.

Performance Objectives: By the end of this lesson, students should be able to:

1. Differentiate between "weather" and "climate."
2. Define "climate change" and "global warming."
3. List the main causes of climate change (greenhouse gases).
4. List three effects of global warming (e.g., sea-level rise, flooding).
5. Differentiate between "mitigation" and "adaptation."
6. Participate in a tree-planting activity.

Lesson Content:

1. Weather vs. Climate

- **Weather:** The short-term state of the atmosphere (e.g., "It is raining *today*," "It is hot *this afternoon*").
- **Climate:** The *average* weather conditions of a region over a *long* period of time (30 years or more). (e.g., "Lagos has a *tropical wet climate*," "Kano has a *hot, dry climate*").

2. Meaning of Climate Change and Global Warming

- **Climate Change:** A significant, long-term change in the average weather patterns of a region (or the whole Earth).
- **Global Warming:** The *long-term heating* of Earth's climate system observed since the pre-industrial period (around 1850-1900) due to human activities. Global warming is the *main cause* of climate change.

3. Causes of Climate Change (The Greenhouse Effect)

1. The Earth is surrounded by a "blanket" of gases (the atmosphere).
 2. Sunlight (shortwave) passes through this blanket and warms the Earth.
 3. The Earth radiates heat (longwave) back out.
 4. **Greenhouse Gases (GHGs)** in the atmosphere (like **Carbon Dioxide - CO₂**) trap this outgoing heat, keeping the planet warm. This is *natural* and good.
 5. **Human Problem:** Since the Industrial Revolution, humans have been burning **fossil fuels** (coal, oil, gas) for factories, cars, and electricity.
 6. This releases *extra* CO₂ into the atmosphere, making the "blanket" thicker.
 7. This "thicker blanket" traps *too much* heat, causing **Global Warming**.
- **Main Causes:**
 - Burning Fossil Fuels (CO₂).
 - **Deforestation:** Trees *absorb* CO₂. When we cut them down, CO₂ increases in the atmosphere.

- Agriculture (Methane from cows).

4. Effects of Global Warming and Climate Change

- **1. Rising Sea Levels:** Ice caps (glaciers) melt, and warm water expands. This is a *huge* threat to coastal cities like Lagos, Port Harcourt, and Calabar, which could be flooded.
- **2. More Extreme Weather:** More intense rainfall and flooding (in the South) and more severe droughts and desertification (in the North).
- **3. Disruption of Agriculture:** Crop failures due to floods or droughts, leading to food insecurity.
- **4. Loss of Biodiversity:** Plants and animals die because they cannot adapt to the new climate.

5. Solutions: Mitigation and Adaptation

- **A. Mitigation (Solving the Problem):** Actions to *reduce* the amount of greenhouse gases in the atmosphere.
 - **Switch to Renewable Energy:** Use solar, wind, and hydro *instead of* fossil fuels.
 - **Afforestation:** Plant massive numbers of trees (trees "eat" CO₂).
 - **Energy Efficiency:** Use less energy (e.g., "Reduce, Reuse, Recycle").
- **B. Adaptation (Living with the Effects):** Actions to *manage* the unavoidable impacts of climate change.
 - *For Flooding:* Build sea walls (like the "Great Wall of Lagos"), improve drainage.
 - *For Drought:* Teach farmers to use drought-resistant crops; build irrigation systems.

6. Practical: School-Based Tree Planting

- The single best *mitigation* strategy a school can do is plant trees.
- Trees fight climate change (absorb CO₂), provide shade (reduce heat), and prevent erosion.
- (Class activity: Students will plant one or two saplings on the school grounds as a practical contribution).

Evaluation:

1. What is the difference between weather and climate?
2. What is global warming?
3. What is the main greenhouse gas, and where does it come from? (CO₂, from burning fossil fuels).
4. List two major effects of climate change in Nigeria.
5. What is the difference between "mitigation" and "adaptation"?

Assignment:

1. Define the "Greenhouse Effect."
2. What is the connection between deforestation and global warming?
3. Give one example of "mitigation" (solving the cause).
4. Give one example of "adaptation" (managing the effect).
5. Write a short paragraph on why your "School Tree Planting" activity was an important scientific project.

The logo for PAWILLS CONSULTING is a large, light blue diamond shape. Inside the diamond is a pink stylized 'P' logo. The words 'PAWILLS' and 'CONSULTING' are written in white capital letters along the bottom-left and bottom-right edges of the diamond, respectively.

WEEK 10: SUSTAINABLE DEVELOPMENT



Content: (I) Meaning and principles (II) Environmental protection (III) SDGs (IV) School eco-project.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Sustainable Development."
2. Explain the "three pillars" of sustainability.
3. Explain the goal of environmental protection in relation to sustainability.
4. Identify the Sustainable Development Goals (SDGs) and list two related to the environment.
5. Plan a simple eco-project for their school.

Lesson Content:

1. Meaning and Principles of Sustainable Development

- **Famous Definition (Brundtland Report, 1987):** "Sustainable Development is development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs."
- **Simpler Meaning:** It means living well today, but not in a way that destroys the planet for our children and grandchildren.
- **Example:**
 - *Not Sustainable:* Cutting down all the trees in a forest to sell the wood today. (The forest is gone, and future generations get nothing).
 - *Sustainable:* Cutting down 100 trees but planting 200 new ones. (You get wood today, and the forest remains for the future).

2. The Three Pillars of Sustainability

For development to be truly sustainable, it must balance three things:

1. **Environmental Sustainability:** Protecting the planet, conserving resources, reducing pollution. (e.g., using solar power).
2. **Economic Sustainability:** Creating jobs and wealth in a way that can last. (e.g., a business that doesn't destroy its own raw materials).
3. **Social Sustainability:** Ensuring a good quality of life for all people (e.g., education, healthcare, human rights).

3. Environmental Protection and Sustainability

Environmental protection is the *core* of sustainability. We must protect our environment so it can continue to provide for us.

- This involves:
 - **Conservation** of resources (Week 3).
 - **Protecting ecosystems** (Week 2).
 - **Managing waste and pollution** (Weeks 4, 8).
 - **Tackling climate change** (Week 10).

4. The Sustainable Development Goals (SDGs)

- The **SDGs** (or "Global Goals") are a set of **17 goals** adopted by all United Nations countries in 2015, to be achieved by 2030. They are a "to-do list" for a sustainable future.
- **Environmental SDGs (Examples):**
 - **Goal 6:** Clean Water and Sanitation
 - **Goal 7:** Affordable and Clean Energy (Solar, Wind)
 - **Goal 11:** Sustainable Cities and Communities
 - **Goal 13:** Climate Action
 - **Goal 14:** Life Below Water
 - **Goal 15:** Life on Land (Stop deforestation)

5. School Eco-Project (Brainstorming) How can our school "meet the needs of the present without compromising the future"?

- **Project Idea 1: Waste Sorting:** Set up labeled bins in the school for "Paper," "Plastic Bottles," and "General Waste."
- **Project Idea 2: Tree Planting:** Plant trees around the school compound to provide shade, clean the air, and prevent erosion.
- **Project Idea 3: "Lights Out" Campaign:** Create posters to remind teachers and students to turn off lights and fans when leaving a classroom (to *reduce* energy use).
- **Project Idea 4: School Garden:** Create a small garden to grow vegetables (sustainable food).

Evaluation:

1. What is the simple definition of sustainable development?
2. What are the "three pillars" of sustainability? (Environmental, Economic, Social).
3. What are the "SDGs"?
4. List two (2) SDGs related to the environment.
5. Suggest one simple eco-project we could do in this school.

Assignment:

1. Write down the famous definition of sustainable development ("...meets the needs of the present...").
2. Explain why cutting down all trees for firewood is *not* sustainable.
3. Explain how planting new trees *is* sustainable.
4. Look up the 17 SDGs. Write down Goal 13, Goal 14, and Goal 15.
5. Write a one-paragraph plan for your "School Eco-Project" (from the lesson).



WEEK 11: REVISION

Topic: Revision

Content: Review of all topics in Term 1.

Note: This week is dedicated to a comprehensive review of all topics covered in the First Term to prepare for the examination.

Revision Checklist:

1. Foundations (Week 1):

- Review of Matter, Energy, Classification.
- Laboratory Safety Rules.

2. Ecosystems (Week 2):

- Biotic (Producers, Consumers, Decomposers) vs. Abiotic (Sun, Water, Soil).
- Interdependence and Balance of Nature.
- Food Chains (one path) vs. Food Webs (many paths).
- Energy flow (starts from Sun, 10% rule).

3. Natural Resources (Week 3):

- Renewable (Sun, Water, Trees) vs. Non-Renewable (Oil, Gas, Coal, Minerals).
- Importance (Revenue, Jobs, Raw Materials).
- Conservation (3 R's: Reduce, Reuse, Recycle; Afforestation).

4. Environmental Hazards (Week 4):

- Drought (lack of rain).
- Desertification (land turning to desert; caused by drought, overgrazing).
- Erosion (soil removal; caused by deforestation, poor farming).
- Flooding (water overflow; caused by rain, blocked drains, building on waterways).

5. Environmental Health & Waste (Weeks 7, 8):

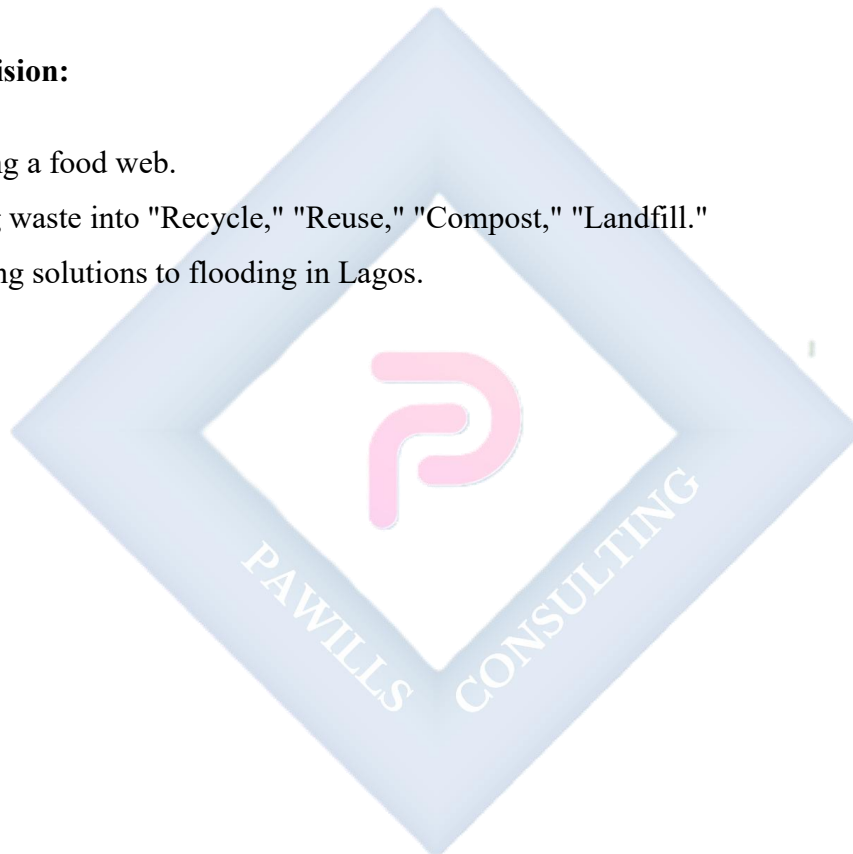
- Environmental Health (clean air, water, waste disposal).
- Role of Individual (don't litter) vs. Government (provide infrastructure).
- Waste Disposal (Landfill, Incineration) vs. Bad Methods (dumping in gutters).

6. Sustainability & Climate (Weeks 9, 10):

- Sustainable Development ("meets needs of present...").
- SDGs (Global Goals).
- Climate Change (long-term change) vs. Weather (short-term).
- Global Warming (heating) caused by Greenhouse Gases (CO₂ from fossil fuels, deforestation).
- Effects (sea-level rise, floods, drought).
- Solutions: Mitigation (stop the cause - plant trees) vs. Adaptation (manage the effects - build sea walls).

Practical Revision:

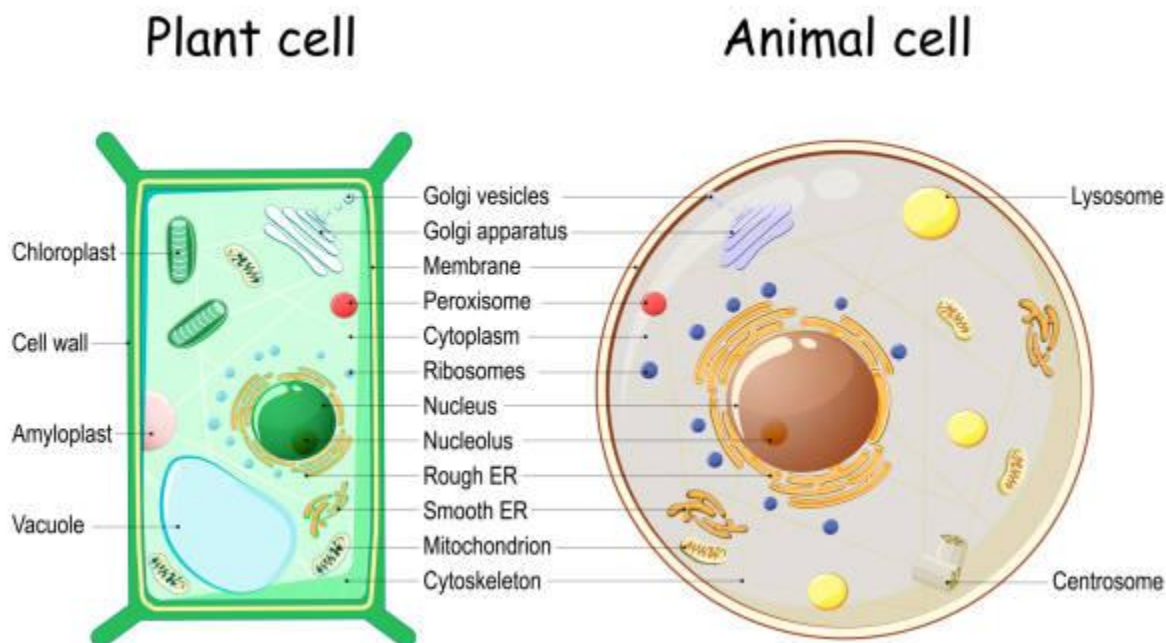
- Drawing a food web.
- Sorting waste into "Recycle," "Reuse," "Compost," "Landfill."
- Debating solutions to flooding in Lagos.



SECOND TERM - JSS 3 INTERMEDIATE SCIENCE LESSON NOTES



WEEK 1: THE CELL AND ITS FUNCTIONS



C

Content: (I) Cell theory (II) Organelles and their roles (III) Differences between plant and animal cells (IV) Observation under microscope.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "cell" as the basic unit of life.
2. State the three (3) main principles of the Cell Theory.
3. Identify and describe the functions of at least six (6) cell organelles (nucleus, cytoplasm, cell membrane, mitochondrion, vacuole, cell wall, chloroplast).
4. List three (3) key differences between a typical plant cell and an animal cell.
5. Demonstrate basic knowledge of how to use a light microscope.

Lesson Content:

1. Definition and The Cell Theory

- **Definition:** The **cell** is the smallest, most basic structural, functional, and biological unit of all known living organisms. It is the "building block" of life. Some organisms (like Amoeba) are made of just one cell (unicellular), while others (like humans) are made of trillions (multicellular).
- **The Cell Theory:** This is the foundational theory of biology, developed by scientists Matthias Schleiden, Theodor Schwann, and Rudolf Virchow. It states three main principles:
 1. All living organisms are composed of one or more cells.
 2. The cell is the basic unit of structure and function in living organisms.
 3. All new cells arise from pre-existing cells (through cell division).

2. Cell Organelles and Their Roles

- **Organelles** ("little organs") are the specialized structures within a cell that perform specific jobs to keep the cell alive.
- **A. Cytoplasm:** The jelly-like substance that fills the cell and surrounds the organelles. It is the site of many chemical reactions.
- **B. Cell Membrane (Plasma Membrane):**
 - **Function:** A thin, flexible barrier that surrounds *all* cells.
 - It is **semi-permeable** (or selectively permeable), meaning it controls what substances (like food, oxygen, and waste) can enter and leave the cell.
- **C. Nucleus:**
 - **Function:** The "control center" or "brain" of the cell.
 - It contains the **chromosomes/DNA** (genetic material), which carry the instructions for all cell activities, including growth and reproduction.
- **D. Mitochondrion (plural: mitochondria):**
 - **Function:** The "powerhouse" of the cell.
 - This is where **respiration** occurs, a process that breaks down food (glucose) to release **energy (ATP)** for the cell to use.
- **E. Ribosomes:**
 - **Function:** The tiny "factories" that build **proteins**. Proteins are needed for growth and repair.

- **F. Vacuole:**

- **Function:** A "storage bag" inside the cell. It stores water, food, and waste.
- **In Plant Cells:** Usually *one, large, central vacuole* that is full of water (cell sap). This water presses against the cell wall (turgor pressure) and keeps the plant firm.
- **In Animal Cells:** Usually *small, numerous, and temporary* vacuoles, if present at all.

- **G. Cell Wall (Plant Cells Only):**

- **Function:** A *tough, rigid* outer layer found *outside* the cell membrane in plants, fungi, and bacteria.
- It is made of **cellulose** (in plants).
- It provides **structural support and protection** to the plant cell, giving it a fixed, rectangular shape.

- **H. Chloroplasts (Plant Cells Only):**

- **Function:** The site of **photosynthesis**.
- They contain a green pigment called **chlorophyll**, which absorbs sunlight.
- This sunlight energy is used to **convert** carbon dioxide and water into glucose (food) for the plant.

3. Differences Between Plant and Animal Cells

Feature	Animal Cell	Plant Cell
Shape	Irregular or round shape	Fixed, rectangular/box-like shape
Cell Wall	Absent	Present (made of cellulose)
Vacuole	Absent , or very small & many	Present (one large central vacuole)
Chloroplasts	Absent (cannot make its own food)	Present (can make its own food)
Food Storage	Stores food as glycogen	Stores food as starch

4. Observation Under a Microscope (Demonstration)

- A **microscope** is an instrument used to see objects that are too small to be seen with the naked eye.

- **Basic Parts:** Eyepiece (to look through), Objective Lens (to magnify), Stage (to place the slide), Coarse/Fine Focus Knobs (to make the image sharp), Light Source (mirror or lamp).
- **Demonstration:** The teacher will show students a prepared slide (e.g., of an onion cell or cheek cell) to demonstrate how the microscope works and to identify the nucleus and cell wall/membrane.

Evaluation:

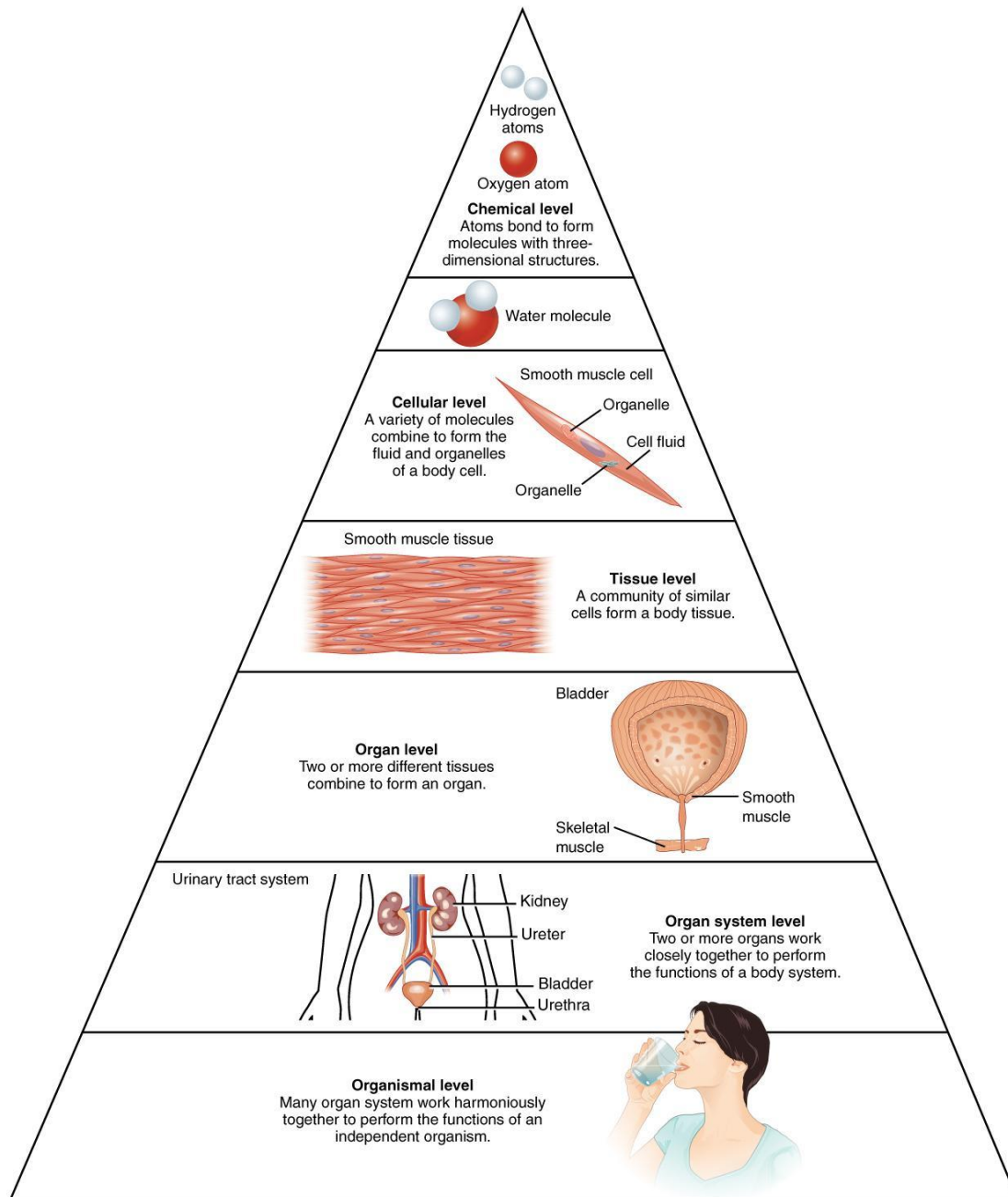
1. What is a cell?
2. State the three (3) principles of the Cell Theory.
3. Which organelle is the "control center" of the cell?
4. Which organelle is the "powerhouse" of the cell?
5. List two (2) differences between a plant cell and an animal cell.

Assignment:

1. Draw a large, neat diagram of a **plant cell**. Label the Cell Wall, Cell Membrane, Cytoplasm, Nucleus, large central Vacuole, and Chloroplast.
2. Draw a large, neat diagram of an **animal cell**. Label the Cell Membrane, Cytoplasm, Nucleus, and Mitochondrion.
3. Create a table listing the functions of: (a) Nucleus, (b) Mitochondrion, (c) Chloroplast, (d) Cell Wall.
4. Why is the cell membrane described as "semi-permeable"?

WEEK 2: TISSUES AND ORGANS

Content: (I) Meaning and examples of tissues (II) Types and functions (III) Organs and organ systems (IV) Importance of coordination.



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "tissue," "organ," and "organ system."
2. Arrange the levels of organization in the correct order (cell → tissue → organ → system → organism).
3. List the four (4) main types of animal tissues and give one example of each.
4. List the main types of plant tissues.
5. Name at least three (3) organ systems and state their main function.

Lesson Content:

1. The Levels of Organization In multicellular organisms (like humans), cells are organized into a hierarchy to perform complex functions.

- **Level 1: Cells:** The basic unit (e.g., muscle cell, nerve cell, skin cell).
- **Level 2: Tissues:** A group of **similar cells** that work together to perform a **specific function**.
- **Level 3: Organs:** A structure made of **different types of tissues** that all work together to perform a complex job.
- **Level 4: Organ Systems:** A group of **different organs** that work together to carry out a major life process.
- **Level 5: Organism:** The complete living being (e.g., a human, a dog, a tree).

Analogy: A school.

- Cell = A single student.
- Tissue = All the JSS3 students (a group of similar units with a specific role).
- Organ = The Science Department (made of different tissues: JSS1, JSS2, JSS3 students, and teachers).
- Organ System = The whole school (made of different organs: Science Dept., Admin, Arts Dept.).
- Organism = The entire Education System.

2. Types of Animal Tissues

1. Epithelial Tissue:

- **Function:** Covers and protects. It forms the *skin* (outer covering) and the *linings* of internal organs (like the stomach and intestines).
- *Example:* Skin, lining of the mouth.

2. Connective Tissue:

- **Function:** Connects, supports, and binds other tissues together.
- *Examples:* **Bone, Cartilage** (in your nose and ears), **Blood** (a liquid connective tissue), tendons, and ligaments.

3. Muscle Tissue:

- **Function:** Contracts and relaxes to produce **movement**.
- *Examples:* Biceps, heart muscle, stomach wall.

4. Nervous Tissue:

- **Function:** Transmits electrical **signals** (messages) throughout the body.
- *Examples:* Found in the **brain, spinal cord, and nerves**.

3. Types of Plant Tissues

1. **Meristematic Tissue:** Found at the tips of roots and shoots; responsible for **growth**.
2. **Dermal Tissue:** The "skin" of the plant (epidermis); protects it.
3. **Vascular Tissue:** The "transport" system.
 - **Xylem:** Transports water *up* from the roots.
 - **Phloem:** Transports food (sugar) *down* from the leaves.
4. **Ground Tissue:** Fills the space in between; site of photosynthesis.

4. Organs and Organ Systems

- **Organ:** A group of different tissues.
 - *Example:* The **Stomach** (Organ) is made of muscle tissue (to churn), epithelial tissue (to line it), and nervous tissue (to control it).
 - *Other Organs:* Heart, Lungs, Brain, Kidney, Leaf (a plant organ).
- **Organ System:** A group of organs working together.

- **Digestive System:** (Organs: Stomach, Small Intestine, Large Intestine, Liver). *Function:* To break down and absorb food.
- **Circulatory System:** (Organs: Heart, Blood Vessels, Blood). *Function:* To transport oxygen and nutrients.
- **Respiratory System:** (Organs: Lungs, Trachea). *Function:* To take in oxygen and remove carbon dioxide.
- **Excretory System:** (Organs: Kidneys, Bladder). *Function:* To remove waste from the blood.
- **Nervous System:** (Organs: Brain, Spinal Cord, Nerves). *Function:* To control and coordinate the body.

5. Importance of Coordination All these systems must work together ("coordinate") to keep the organism alive.

- *Example:* When you run (Muscular System), your Lungs (Respiratory System) work faster to get oxygen, and your Heart (Circulatory System) works faster to pump that oxygen to the muscles. This is all controlled by your Brain (Nervous System).

Evaluation:

1. What is a tissue?
2. What is an organ?
3. List the 5 levels of organization in order, from smallest to largest.
4. Name the four (4) main types of animal tissues.
5. What is the function of the Digestive System? Name one organ in it.

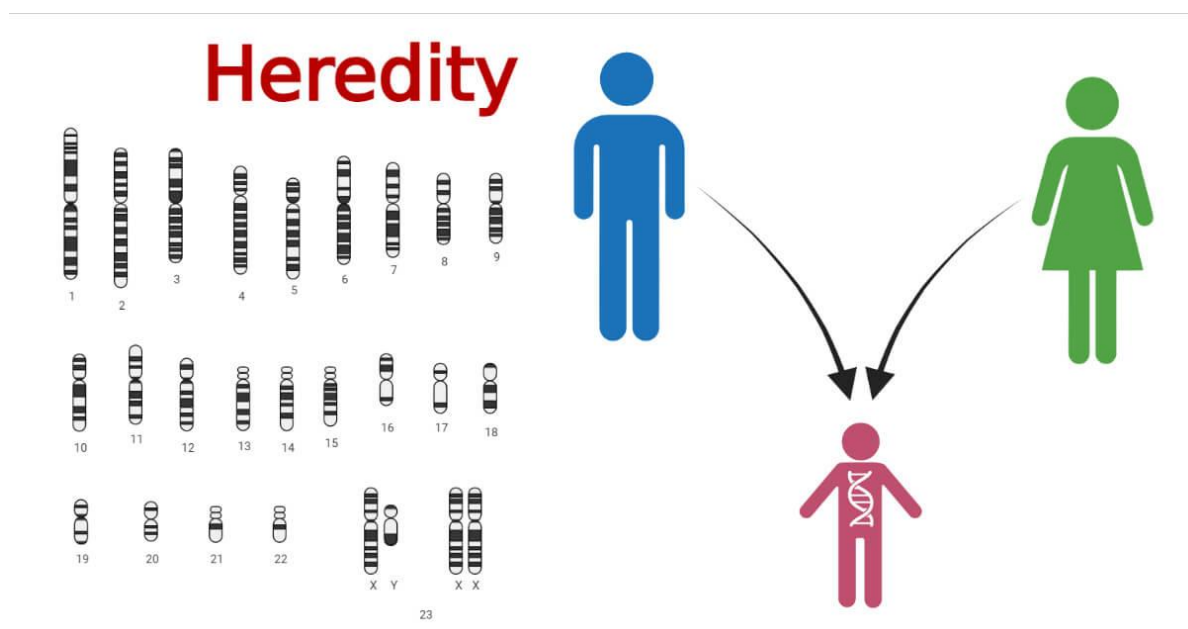
Assignment:

1. Define Tissue, Organ, and Organ System.
2. For each of the four animal tissues, state its main function and give one example.
3. Name the two transport tissues in plants and state what each one transports.
4. Choose two (2) human organ systems. For each, list its main function and two (2) organs it contains.

5. Explain why "coordination" between organ systems is essential for survival.



WEEK 3: HEREDITY AND VARIATION



Content: (I) Meaning of heredity (II) Inherited and acquired traits (III) Importance of variation (IV) Simple genetic inheritance.

Performance Objectives: By the end of this lesson, students should be able to:

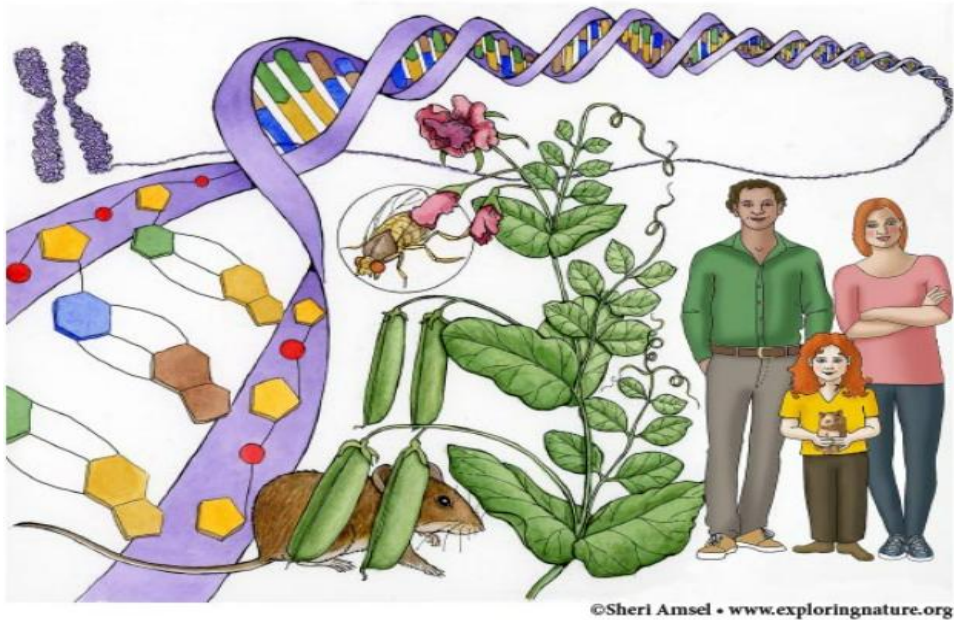
1. Define Heredity, Genetics, Trait, Inherited Trait, and Acquired Trait.
2. Give at least three examples of inherited traits and three examples of acquired traits.
3. Define Variation and explain its importance for survival (adaptation).
4. Explain the basic concepts of dominant and recessive genes.

Lesson Content:

1. Meaning of Heredity

- **Heredity:** The passing of **traits** (characteristics) from **parents** to their **offspring** (children). It is the reason why children look like their parents.
- **Genetics:** The *scientific study* of heredity and how traits are passed on.

- **Genes:** The units of heredity. They are segments of **DNA** (Deoxyribonucleic Acid) found on **chromosomes** inside the nucleus of our cells. Genes are the "instruction manual" that determines our traits.



2. Inherited vs. Acquired Traits

- **A. Inherited Traits:**
 - **Definition:** Characteristics that you are **born with**. They are passed down from your parents through genes. You have *no control* over them.
 - **Examples:**
 - Blood group (A, B, AB, O)
 - Skin colour
 - Eye colour
 - Natural hair colour and texture (curly/straight)
 - Ability to roll your tongue
 - Attached or free earlobes
- **B. Acquired Traits:**

- **Definition:** Characteristics that you "get" or *develop* during your lifetime. They are *not* in your genes and **cannot** be passed on to your children.
- **Examples:**
 - Scars
 - Muscles from exercise
 - Ability to speak a language (e.g., English, Yoruba)
 - Knowledge from school
 - A tattoo or tribal mark
 - Bleached skin

3. Variation

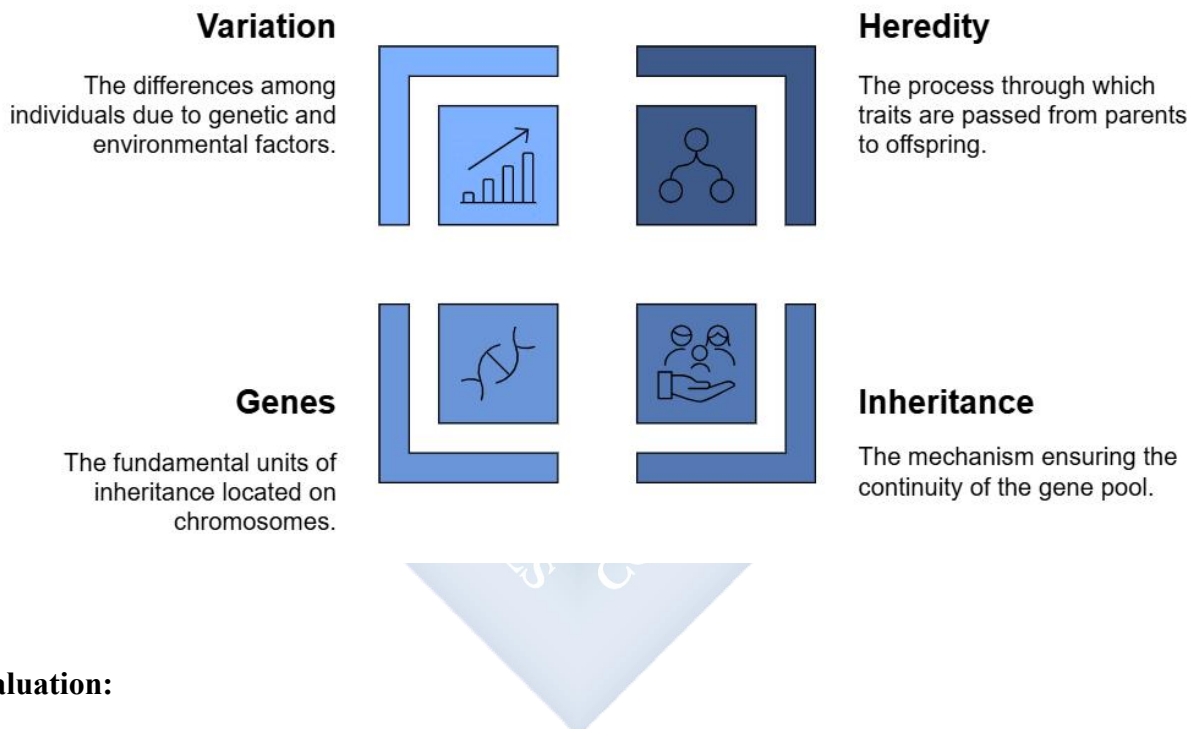
- **Definition:** The **differences** (in traits) that exist among individuals of the *same species*. It is why everyone in your class (except identical twins) looks different from each other.
- **Causes of Variation:**
 1. **Sexual Reproduction:** The *mixing* of genes from two different parents (a father and a mother) creates new, unique combinations.
 2. **Environment:** Your environment can also cause differences (e.g., two plants with the same genes, but one gets more sun and grows taller).
- **Importance of Variation:**
 - Variation is the raw material for **evolution** and **adaptation**.
 - It allows a species to **survive** changes in the environment.
 - *Example:* In a population of goats, some have thick fur, and some have thin fur (variation). If the climate suddenly becomes very cold, the thin-furred goats might die, but the thick-furred goats (thanks to variation) will survive, reproduce, and pass on the "thick fur" gene.

4. Simple Genetic Inheritance (Mendelian Genetics)

- For each trait, you get two genes (one from your father, one from your mother).
- **Dominant Gene:** The "stronger" gene. If it is present, its trait will *always* show. (Represented by a capital letter, e.g., **T** for Tall).

- **Recessive Gene:** The "weaker" gene. Its trait is *hidden* if a dominant gene is present. It will only show if *both* genes are recessive. (Represented by a small letter, e.g., **t** for short).
- **Example (Height):**
 - If a plant gets **TT** (from both parents) → It is **Tall**.
 - If a plant gets **tt** (from both parents) → It is **short**.
 - If a plant gets **Tt** (one of each) → It is **Tall** (because the **T** is dominant and *hides* the **t**).

Genetics



Evaluation:

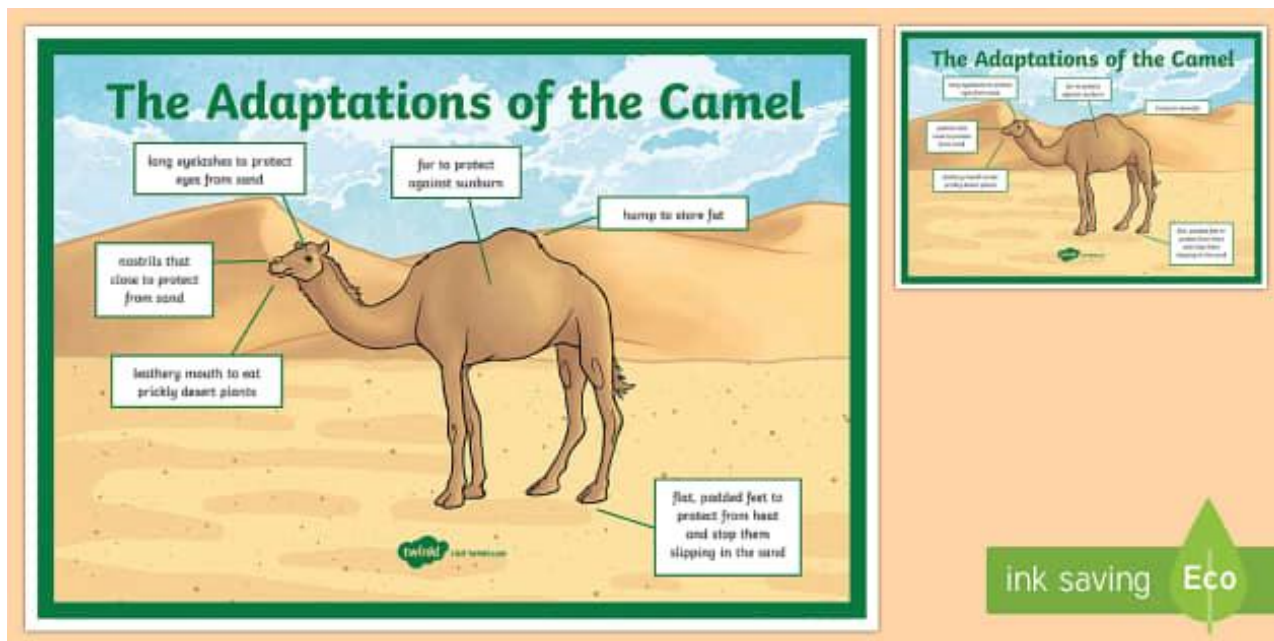
1. What is heredity?
2. What is the difference between an inherited trait and an acquired trait?
3. Is "ability to speak French" an inherited or acquired trait? Why?
4. What is variation?
5. What is a dominant gene?

Assignment:

1. Define Heredity, Gene, and Trait.
2. List three (3) inherited traits and three (3) acquired traits.
3. Explain, in your own words, why variation is important for the survival of a species.
4. My father has free earlobes (a dominant trait, **E**) and my mother has attached earlobes (a recessive trait, **e**). If my father's genes are **Ee**, what is the *percentage chance* I will have attached earlobes? (Draw a Punnett square. Answer: 50%).
5. What is the name of the acid that carries genetic information?



WEEK 4: ADAPTATION OF ORGANISMS



Content: (I) Meaning and examples (II) Adaptation to habitat (III) Structural and behavioral adaptations (IV) Importance of adaptation.

Performance Objectives: By the end of this lesson, students should be able to:

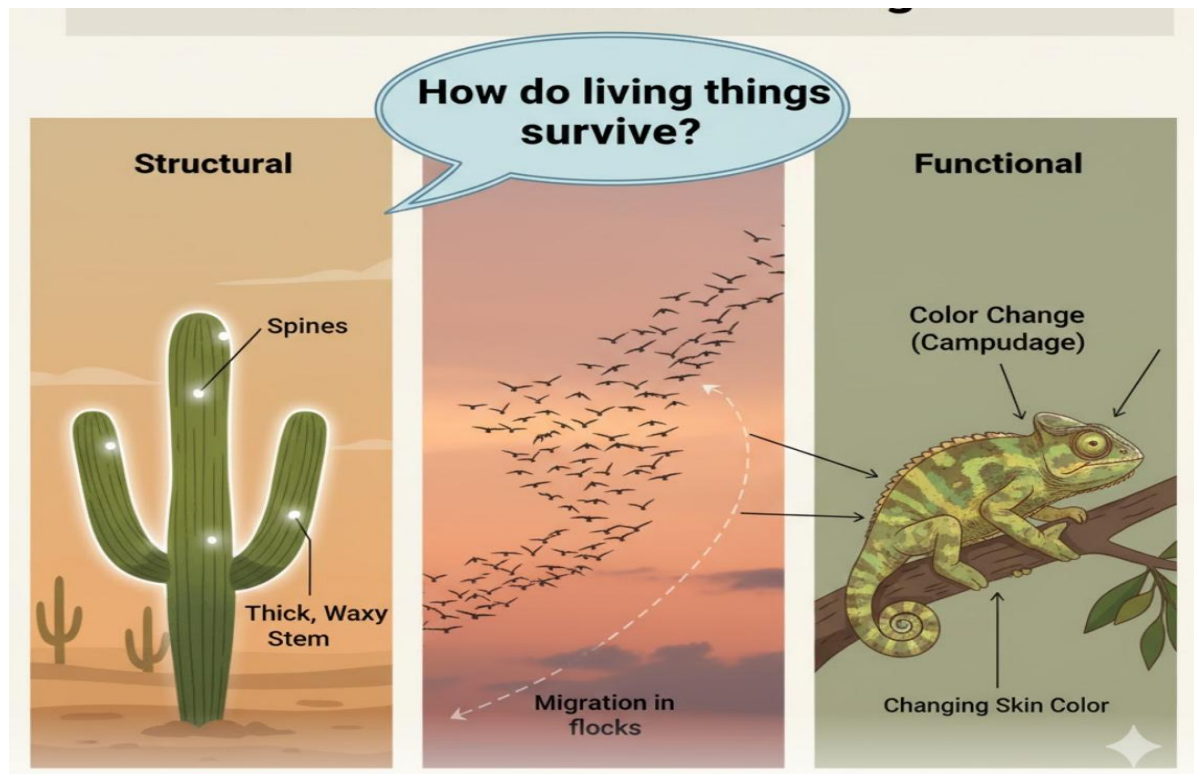
1. Define "Adaptation" and "Habitat."
2. Differentiate between structural, behavioral, and functional adaptations.
3. Provide at least two examples of structural adaptation (e.g., camel's hump, fish's gills).
4. Provide at least one example of behavioral adaptation (e.g., migration, hibernation).
5. Explain how adaptation is essential for survival.

Lesson Content:

1. Meaning of Adaptation and Habitat

- **Habitat:** The *natural home* or environment where an organism lives (e.g., the desert, a river, a forest).
- **Adaptation:** A special **trait** or **feature** (a change in the body or behavior) that helps an organism to **survive** and **reproduce** successfully in its specific habitat.

- **Importance:** Organisms that are *well-adapted* to their environment survive, find food, and have offspring. Organisms that are *poorly-adapted* will die.

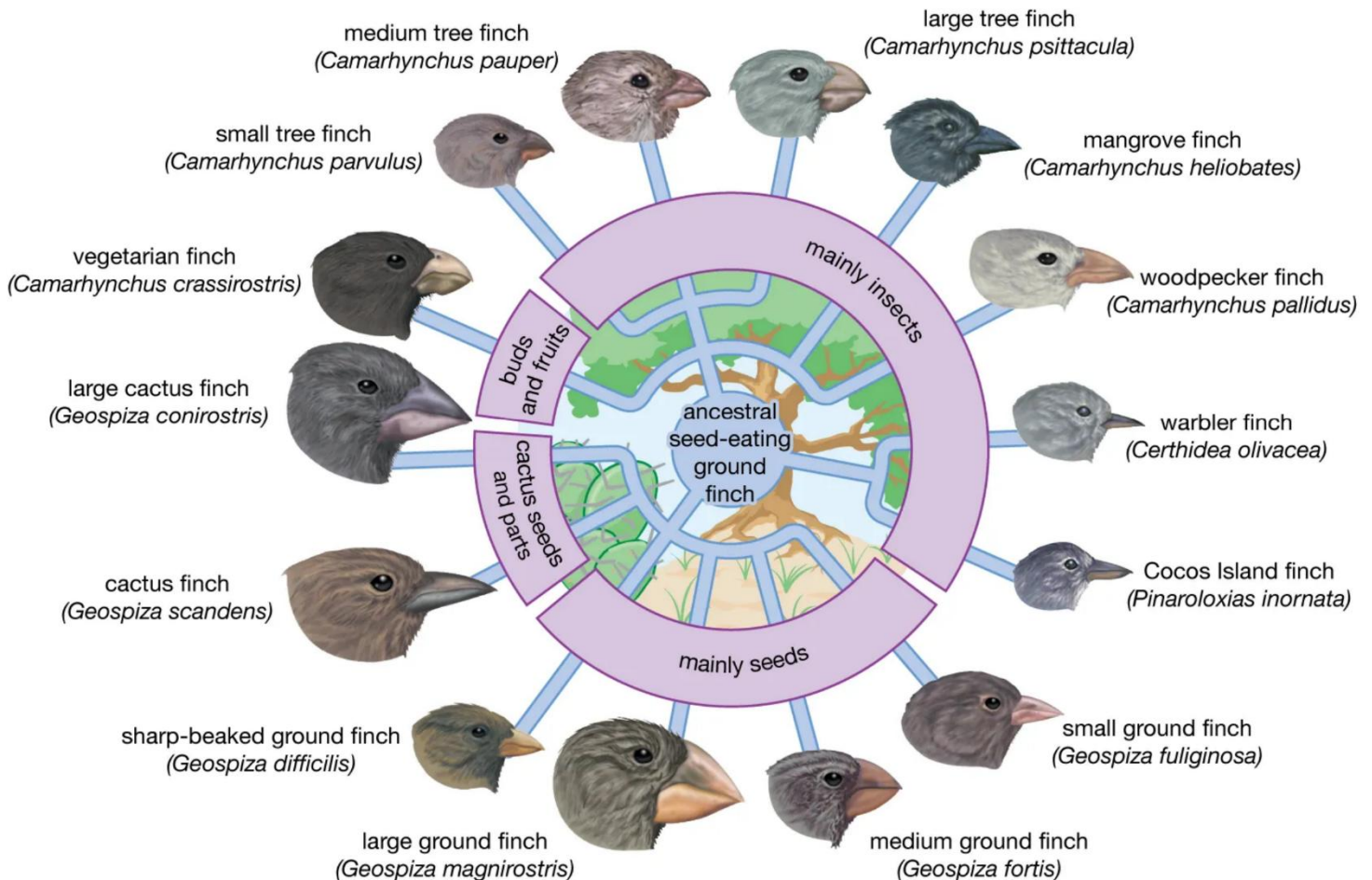


2. Types of Adaptations

- **A. Structural (or Physical) Adaptations:**
 - **Definition:** These are *physical features* of an organism's **body** that help it survive.
 - **Examples:**
 1. **Camel (in the desert):**
 - **Hump:** Stores fat (not water), which can be broken down for energy and water.
 - **Long Eyelashes:** Keep sand out of its eyes.
 - **Wide, flat feet:** To stop it from sinking in the sand.
 2. **Fish (in water):**
 - **Gills:** To breathe oxygen dissolved in water.

- **Fins and Streamlined Body:** To swim fast.
 - **Scales:** For protection.
3. **Cactus (in the desert):**
- **Spines (needles) instead of leaves:** To reduce water loss and protect from animals.
 - **Thick, fleshy stem:** To store large amounts of water.
4. **Polar Bear (in the Arctic):**
- **Thick white fur:** For camouflage (hiding) and warmth.
 - **Thick layer of blubber (fat):** For insulation against the cold.

Adaptive radiation in Galapagos finches



- **B. Behavioral Adaptations:**

- **Definition:** These are the *actions* or *things an organism does* to help it survive.
- **Examples:**
 1. **Migration:** Birds flying south (to a warmer place) for the winter to find food.
 2. **Hibernation:** Bears sleeping through the entire winter to save energy when food is scarce.
 3. **Nocturnal Activity:** Animals (like owls or rats) sleeping during the hot day and coming out at night.
 4. **Living in Groups:** Lions hunting in a pride, or meerkats posting a "lookout" for predators.

- **C. Functional (or Physiological) Adaptations:**

- **Definition:** These are *internal processes* or *chemical functions* of an organism's body.
- **Examples:**
 1. **Poison/Venom:** Snakes producing venom to kill prey; plants producing poison to stop herbivores.
 2. **Color Change:** A chameleon changing its skin colour to match its surroundings (camouflage).

Evaluation:

1. What is an adaptation?
2. What is a habitat?
3. What is the difference between a structural and a behavioral adaptation?
4. A fish's gills are what type of adaptation? (Structural).
5. A bird flying south for the winter is what type of adaptation? (Behavioral).

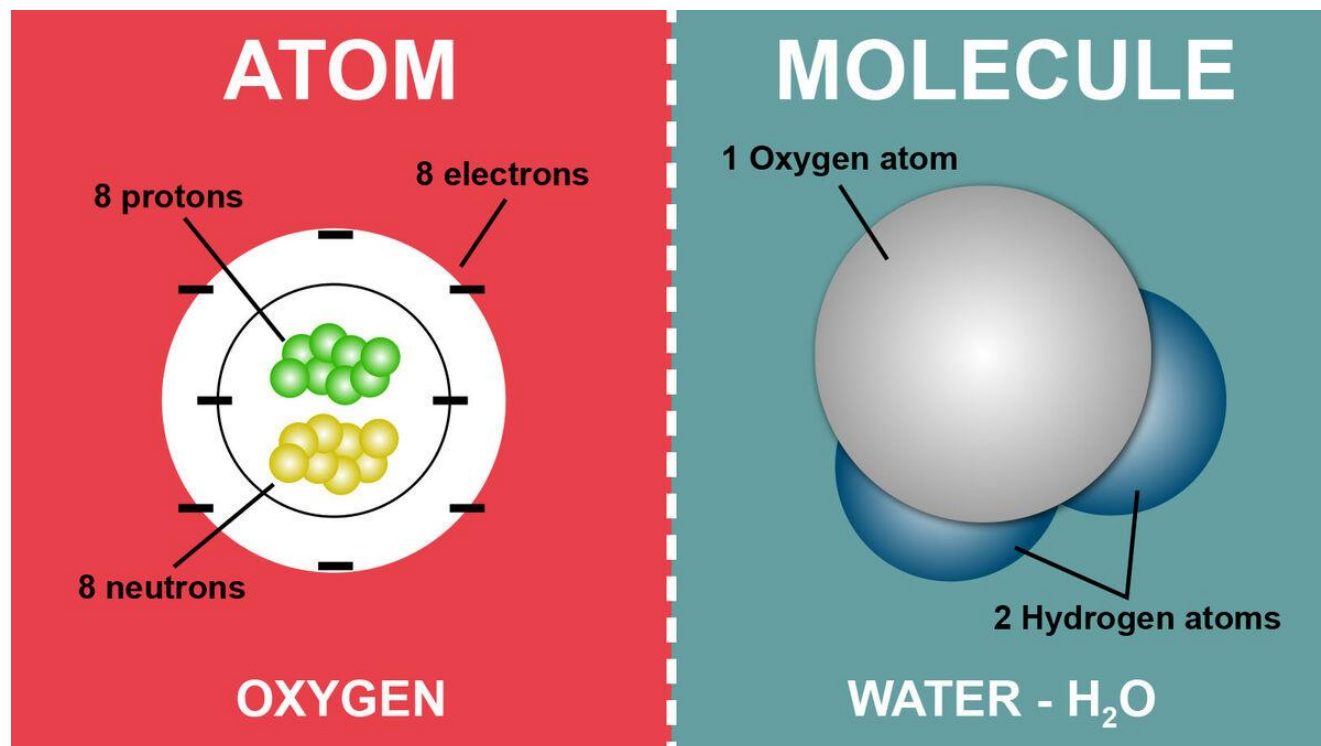
Assignment:

1. Define Adaptation and Habitat.
2. List and define the three types of adaptation (Structural, Behavioral, Functional).

3. Choose one habitat (e.g., desert, river, rainforest).
4. Describe one (1) plant and one (1) animal that lives there.
5. For *each* of them, list two (2) adaptations they have and explain how it helps them survive.



WEEK 6: ATOMS AND MOLECULES



Content: (I) Basic structure of atom (II) Elements, compounds, and mixtures (III) Symbols and formulae (IV) Simple chemical equations.

Performance Objectives: By the end of this lesson, students should be able to:

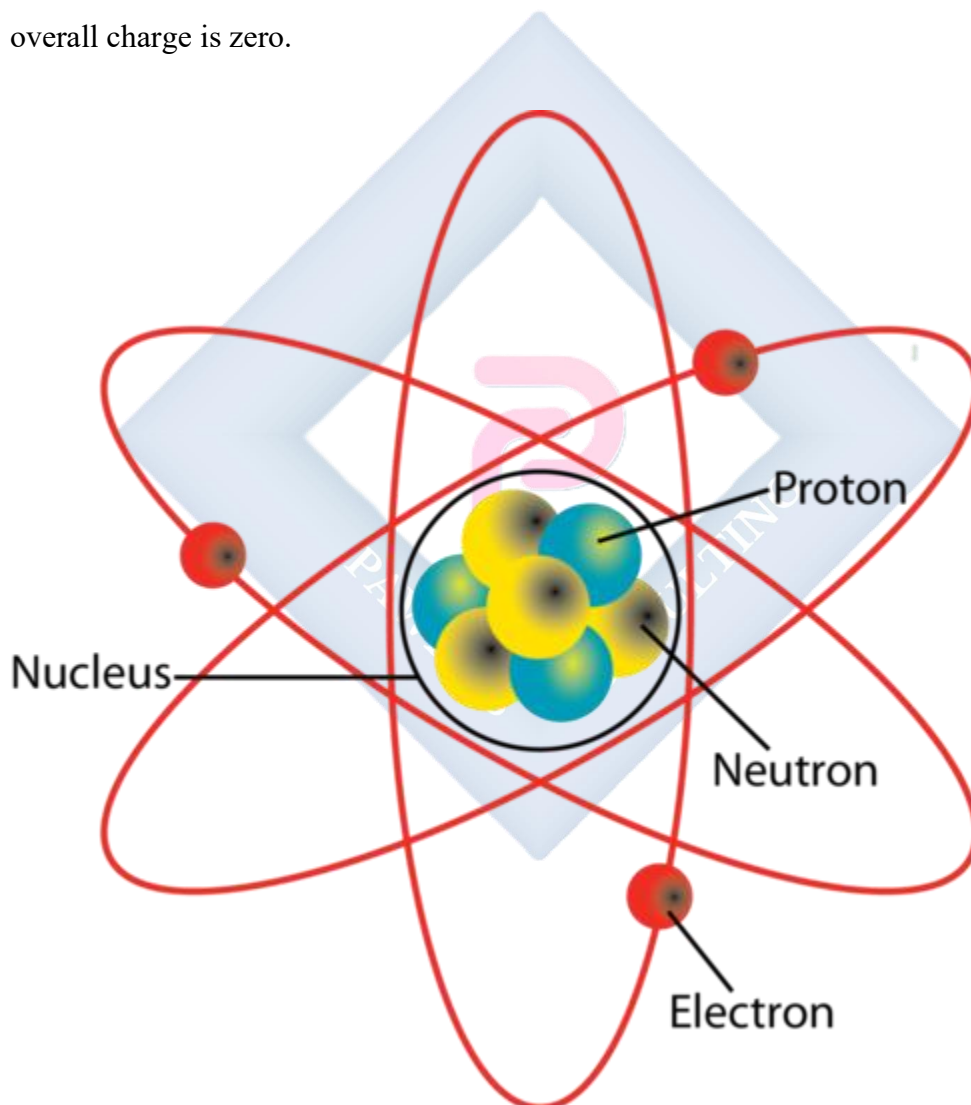
1. Define "atom" as the building block of matter.
2. Draw and label the basic structure of an atom (proton, neutron, electron).
3. Define element, compound, and mixture (review).
4. Write the chemical symbols for the first 20 elements.
5. Write the chemical formulae for simple compounds (e.g., H₂O, CO₂, NaCl).
6. Understand (but not yet balance) a simple chemical equation.

Lesson Content:

1. Basic Structure of an Atom

- **Definition:** An **atom** is the smallest particle of an element that can take part in a chemical reaction. It is the basic "building block" of all matter.

- **Structure:** An atom is made of two main parts:
 1. **The Nucleus:** The tiny, dense, *center* of the atom. It contains two particles:
 - **Protons (p^+):** Positively charged particles (+).
 - **Neutrons (n^0):** Neutral particles (no charge).
 2. **The Electron Shells (or Orbits):** The "paths" *outside* the nucleus. They contain:
 - **Electrons (e^-):** Negatively charged particles (-). They orbit the nucleus at high speed.
- In a neutral atom, the number of protons (+) is *equal* to the number of electrons (-), so the overall charge is zero.



2. Elements, Compounds, and Mixtures (Review)

- **Element:** A pure substance made of *only one type of atom*. (e.g., Gold (Au), Oxygen (O)).

- **Compound:** A pure substance made of *two or more different types of atoms chemically bonded together* in a fixed ratio. (e.g., Water (H_2O) - always 2 Hydrogen atoms, 1 Oxygen atom).
- **Molecule:** The smallest particle of an element or compound that can exist on its own. (e.g., a molecule of Oxygen, O_2 ; a molecule of Water, H_2O).
- **Mixture:** Two or more substances (elements or compounds) *physically mixed* together, not chemically bonded. (e.g., Salt water).

Chemistry compound formula				
H_2O Water	NaCl Sodium Chloride	CuSO_4 Copper Sulphate	H_2SO_4 Sulphuric Acid	HNO_3 Nitric Acid
NaOH Sodium Hydroxide	SO_2 Sulphur dioxide	CO_2 Carbon dioxide	MgO Magnesium oxide	KCl Potassium Chloride
CO_2 Carbon dioxide	MgCl_2 Magnesium Chloride	H_2S Hydrogen Sulphide	CH_4 Methane	HCl Hydrochloric acid
NH_3 Ammonia Gas	NaCl Sodium Chloride	C_2H_6 Ethane	C_3H_8 Propane	C_4H_{10} Butane

3. Chemical Symbols and Formulae

- **Chemical Symbol:** A 1- or 2-letter "shorthand" for an element. The first letter is *always* capital; the second (if any) is *always* small.

○ **First 20 Elements (to memorize):**

1. H - Hydrogen
2. He - Helium
3. Li - Lithium
4. Be - Beryllium
5. B - Boron
6. C - Carbon
7. N - Nitrogen
8. O - Oxygen
9. F - Fluorine
10. Ne - Neon
11. Na - Sodium (from Natrium)
12. Mg - Magnesium
13. Al - Aluminum
14. Si - Silicon
15. P - Phosphorus
16. S - Sulfur
17. Cl - Chlorine
18. Ar - Argon
19. K - Potassium (from Kalium)
20. Ca - Calcium

- **Chemical Formula:** A "shorthand" for a compound. It shows *which* elements and *how many* atoms of each are in one molecule.
 - **Water (H₂O):** 2 atoms of Hydrogen, 1 atom of Oxygen.
 - **Carbon Dioxide (CO₂):** 1 atom of Carbon, 2 atoms of Oxygen.
 - **Table Salt (NaCl):** 1 atom of Sodium, 1 atom of Chlorine.
 - **Glucose (C₆H₁₂O₆):** 6 Carbon, 12 Hydrogen, 6 Oxygen.

4. Simple Chemical Equations

- **Definition:** A chemical equation is a "sentence" that describes a chemical reaction.
- It shows the **Reactants** (what you start with) and the **Products** (what you end with).

- Reactants → Products
- **Example 1 (Burning Carbon):**
 - *Word Equation:* Carbon + Oxygen → Carbon Dioxide
 - *Chemical Equation:* $C + O_2 \rightarrow CO_2$
- **Example 2 (Making Water):**
 - *Word Equation:* Hydrogen + Oxygen → Water
 - *Chemical Equation:* $2H_2 + O_2 \rightarrow 2H_2O$ (This is "balanced" - we have 4 H and 2 O on both sides).

Evaluation:

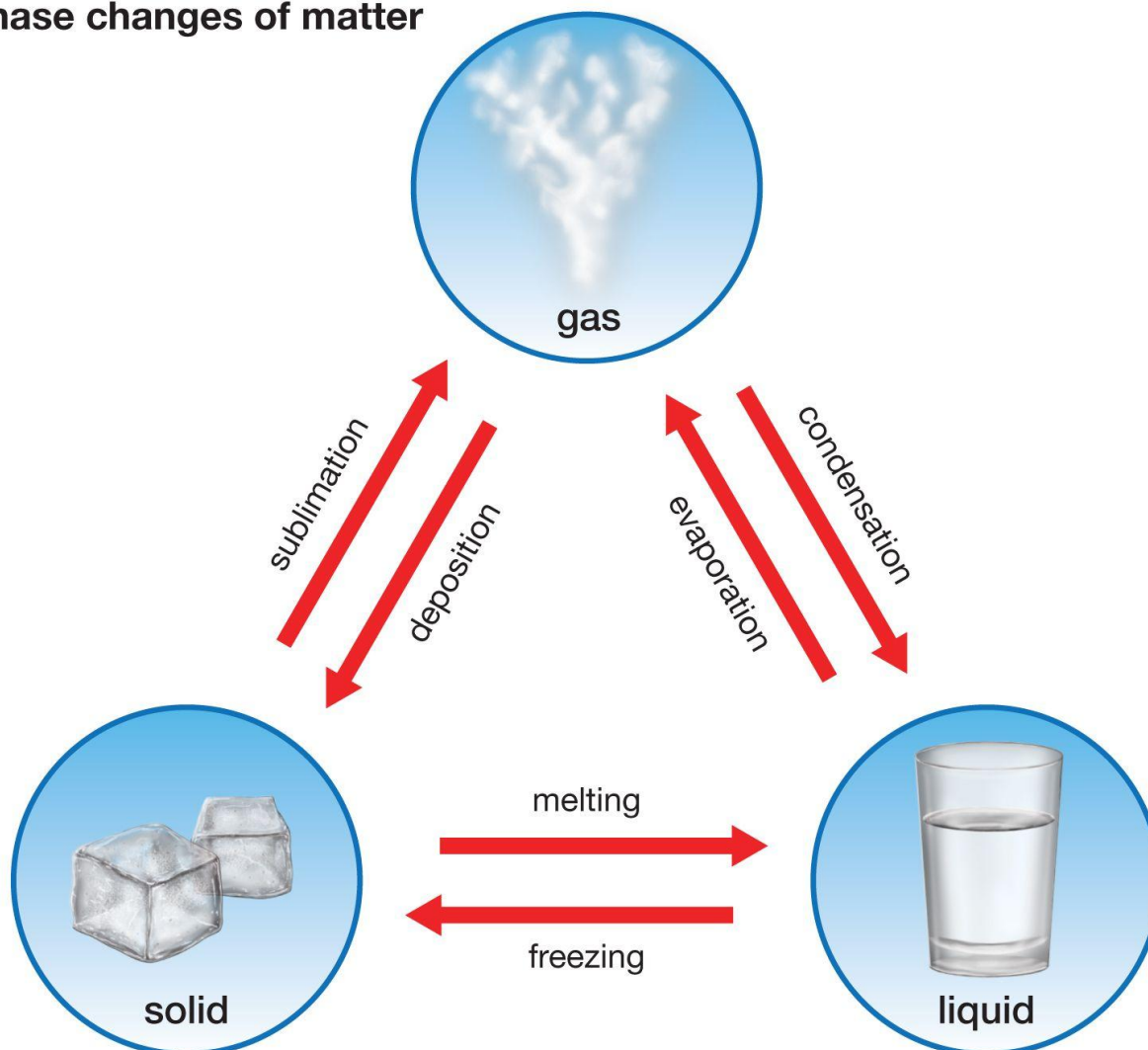
1. What is an atom?
2. What are the three sub-atomic particles, and what is the charge of each?
3. What is the difference between an element and a compound?
4. What is the chemical symbol for Sodium?
5. What does the formula CO_2 tell us?

Assignment:

1. Draw and label the basic structure of a Helium atom (2 protons, 2 neutrons, 2 electrons).
2. Memorize and write down the chemical symbols for the first 10 elements.
3. Write the chemical formulae for: (a) Table Salt, (b) Water, (c) Methane (1 Carbon, 4 Hydrogen).
4. Write the *word equation* for the reaction between Carbon and Oxygen.
5. What is the difference between a chemical symbol and a chemical formula?

WEEK 8: CHANGES IN MATTER

Phase changes of matter



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Content: (I) Physical and chemical changes (II) Reversible and irreversible processes (III) Simple experiments.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define and differentiate between a physical change and a chemical change.
2. Provide at least two examples for each type of change.
3. Define reversible and irreversible changes.

4. Correctly classify common processes (e.g., melting, burning, rusting) as physical or chemical, reversible or irreversible.
5. Observe and describe a simple chemical change.

Lesson Content:

1. Physical Changes

- **Definition:** A **physical change** is a change that affects the *form* or *appearance* of a substance, but **does not** change its *chemical identity*.
- **Key Features:**
 1. **No new substance is formed.** (Water is still H₂O, whether it is ice, liquid, or steam).
 2. The change is usually **reversible**.
 3. No new energy is produced (or very little).
- **Examples:**
 - **Change of State:**
 - Melting (solid to liquid) - e.g., ice melting.
 - Freezing (liquid to solid) - e.g., water turning to ice.
 - Evaporation (liquid to gas) - e.g., boiling water.
 - Condensation (gas to liquid) - e.g., water on a cold glass.
 - **Dissolving:** Salt or sugar dissolving in water. (You can get the salt back by evaporating the water).
 - **Changing Shape:** Cutting paper, crushing a can, breaking glass.

2. Chemical Changes

- **Definition:** A **chemical change** (or chemical reaction) is a change that results in the formation of **one or more new substances** with *different properties*.
- **Key Features:**
 1. **A new substance is formed.** (Ash is not the same as wood).
 2. The change is usually **irreversible** (you can't "un-burn" wood).

3. Energy (heat, light, sound) is often produced or absorbed.

- **Examples:**

- **Burning (Combustion):** Burning wood, paper, or gas. (Produces ash, carbon dioxide, etc.).
- **Rusting:** Iron + Oxygen $\rightarrow$ Iron Oxide (Rust).
- **Cooking/Baking:** An egg, once fried, cannot be made raw again.
- **Digestion:** The food in your stomach is broken down by acids into new chemicals.
- **Decay/Rotting:** Food or leaves rotting.
- **Reaction with Acid:** (See experiment below).

Chemical and Physical Changes

Chemical change: A chemical reaction forms new products.

Physical change: Matter changes form but not chemical identity.

			
Combustion	Rotting	Melting	Shredding
			
Rusting	Digestion	Boiling	Chopping

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3. Reversible vs. Irreversible

- **Reversible Change:** A change that can be easily "undone" to get the original materials back. (Most physical changes).

- *Example:* Melting ice. You can freeze it back.
- **Irreversible Change:** A change that cannot be undone. (Most chemical changes).
 - *Example:* Frying an egg. You cannot make it raw again.

4. Simple Experiment (Chemical Change)

- **Title:** Reaction of an acid with a carbonate.
- **Materials:** Vinegar (an acid), baking soda (a carbonate), a beaker or cup.
- **Procedure:** Pour some vinegar into the cup. Add a spoonful of baking soda.
- **Observation:** The mixture fizzes, bubbles, and produces a gas.
- **Conclusion:** This is a **chemical change**. A new substance (carbon dioxide gas) was produced, and the change is irreversible.

Evaluation:

1. What is a physical change? Give one example.
2. What is a chemical change? Give one example.
3. Is *burning* paper a physical or chemical change? Why?
4. Is *melting* ice a physical or chemical change? Why?
5. What is the difference between a reversible and an irreversible change?

Assignment: Classify the following changes as either **Physical** or **Chemical**.

1. Rusting of a nail.
2. Boiling water.
3. Burning a candle.
4. Dissolving sugar in tea.
5. Baking a cake.
6. Cutting a piece of cloth.
7. Digesting food.
8. Freezing water into ice cubes.

WEEK 9: WATER AND AIR



Content: (I) Composition and importance (II) Water purification and conservation (III) Air pressure and experiments.

Performance Objectives: By the end of this lesson, students should be able to:

1. State the chemical formula for water and list the main components of air.
2. List three (3) important uses of water and two (2) important uses of air.
3. Describe the steps in municipal water purification.
4. Define "air pressure" and describe an experiment to show its existence.

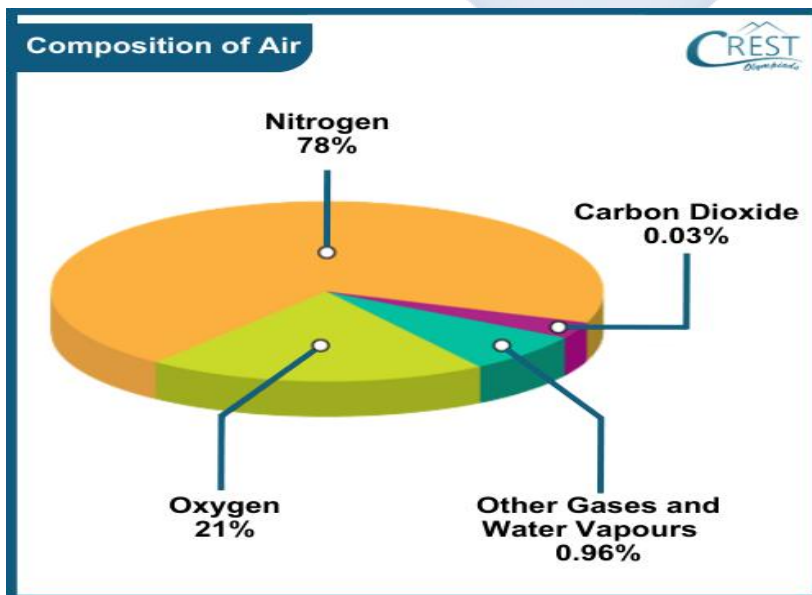
Lesson Content:

1. Water (H₂O)

- **Composition:** A compound made of 2 atoms of Hydrogen and 1 atom of Oxygen (H₂O).
- **Importance (Uses):**

1. **Drinking:** Essential for all life; for digestion and blood circulation.
 2. **Domestic:** Cooking, washing, bathing, cleaning.
 3. **Agriculture:** Irrigation for crops.
 4. **Industrial:** Used as a coolant, a solvent, and for generating electricity (hydro-electric power).
- **Water Purification:** The process of removing dirt, chemicals, and germs from water to make it safe to drink (potable).
 - **Simple Methods:** Boiling, Filtration.
 - **Municipal Water Treatment (Large Scale):**
 1. **Screening:** Removes large objects (sticks, leaves).
 2. **Coagulation/Flocculation:** Chemicals (like Alum) are added to make tiny dirt particles clump together (floc).
 3. **Sedimentation:** The heavy "floc" settles to the bottom of the tank.
 4. **Filtration:** The clear water on top is passed through layers of sand and gravel to remove fine particles.
 5. **Disinfection:** A chemical (like **Chlorine**) is added to kill any remaining bacteria and viruses.
 - **Water Conservation:** Using water wisely. (e.g., turning off taps, fixing leaks, recycling water).

2. Air (A Mixture)



- **Composition:** Air is a **mixture** of gases. Its approximate composition is:

- **Nitrogen (N₂):** 78% (Most abundant)
- **Oxygen (O₂):** 21% (Most important for life)
- **Argon (Ar):** 0.9%
- **Carbon Dioxide (CO₂):** 0.04%
- *Variable amounts of:* Water Vapour, dust, etc.

- **Importance (Uses):**

1. **Oxygen:** Essential for **respiration** (breathing) in all animals.
2. **Carbon Dioxide:** Essential for **photosynthesis** in plants.
3. **Nitrogen:** A key component of proteins (for growth).
4. **Air (Wind):** Used to generate electricity (wind turbines), for sailing, etc.

3. Air Pressure

- **Definition:** Air pressure is the **weight** of the air in the atmosphere pressing down on the Earth's surface. We don't feel it because it presses on us equally from all directions.

- **Experiment 1: The Crushing Can**

1. Put a small amount of water in an empty aluminum can.
2. Heat the can until the water boils and steam comes out (this pushes all the air *out*).
3. Quickly turn the can upside-down and plunge it into a bowl of cold water.
4. **Observation:** The can will be instantly crushed.
5. **Explanation:** The steam inside cooled, condensed, and created a vacuum (low pressure). The *high* air pressure *outside* the can was then strong enough to crush it.

- **Experiment 2: The Inverted Glass**

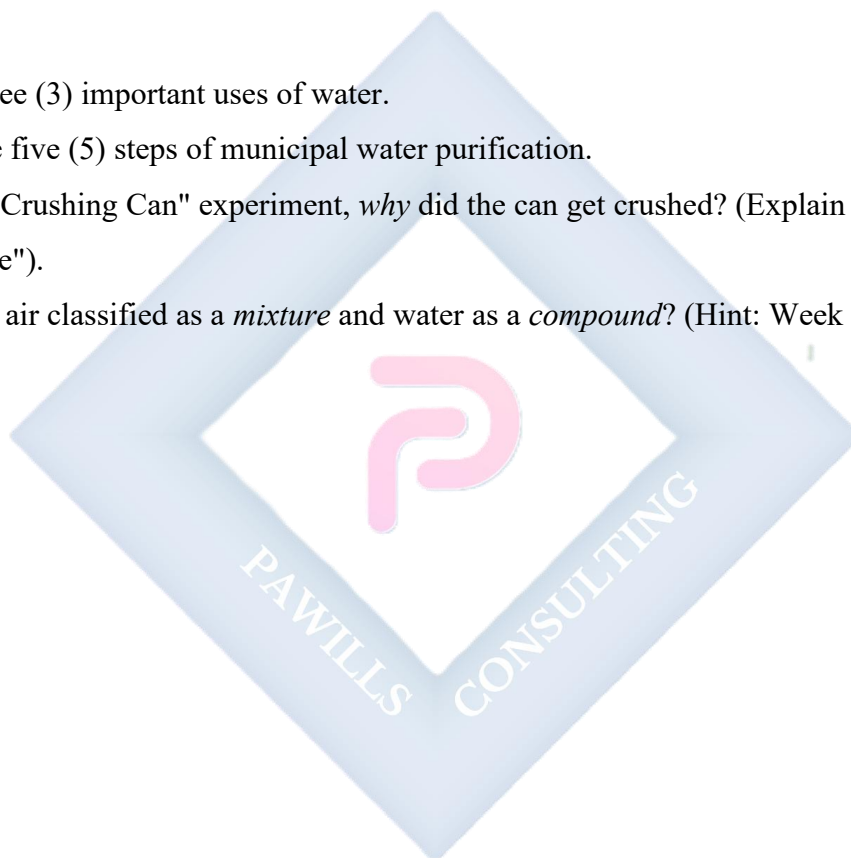
1. Fill a glass to the brim with water.
2. Place a flat piece of cardboard over the top.
3. Holding the cardboard, turn the glass upside-down.
4. **Observation:** The cardboard stays in place, and the water does not fall out.
5. **Explanation:** The force of the *air pressure pushing up* on the cardboard is stronger than the *weight* of the water *pushing down*.

Evaluation:

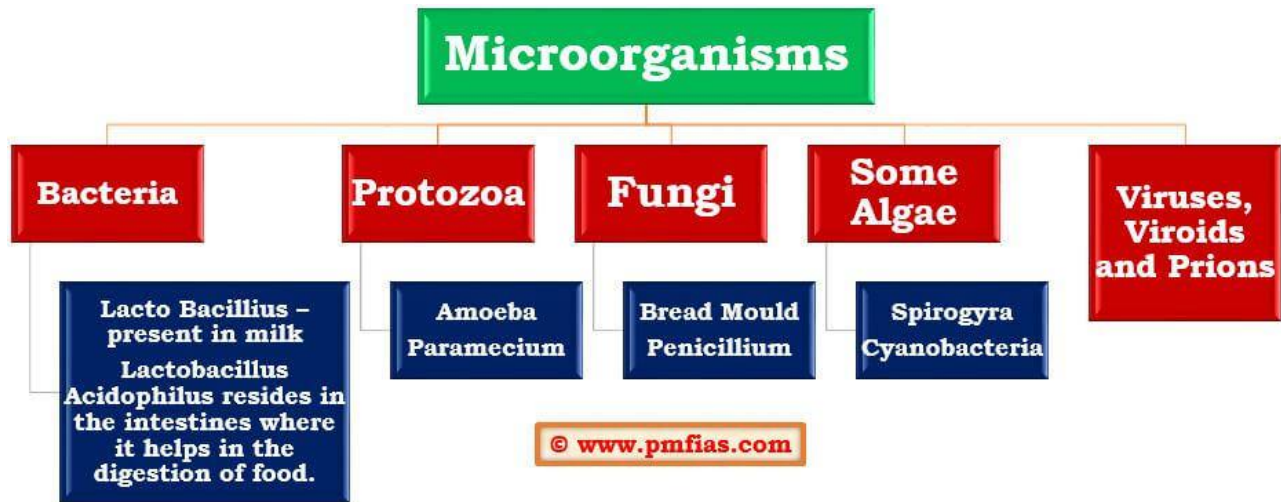
1. What is the chemical formula for water?
2. What is the most abundant gas in the air? (Nitrogen).
3. What is the most important gas for respiration? (Oxygen).
4. What chemical is added at the *end* of water purification to kill germs? (Chlorine).
5. What is air pressure?

Assignment:

1. List three (3) important uses of water.
2. List the five (5) steps of municipal water purification.
3. In the "Crushing Can" experiment, *why* did the can get crushed? (Explain using "air pressure").
4. Why is air classified as a *mixture* and water as a *compound*? (Hint: Week 1).



WEEK 10: MICROORGANISMS AND DISEASES



Content: (I) Pathogens and disease transmission (II) Prevention and control (III) Vaccination and hygiene practices.

Performance Objectives: By the end of this lesson, students should be able to:

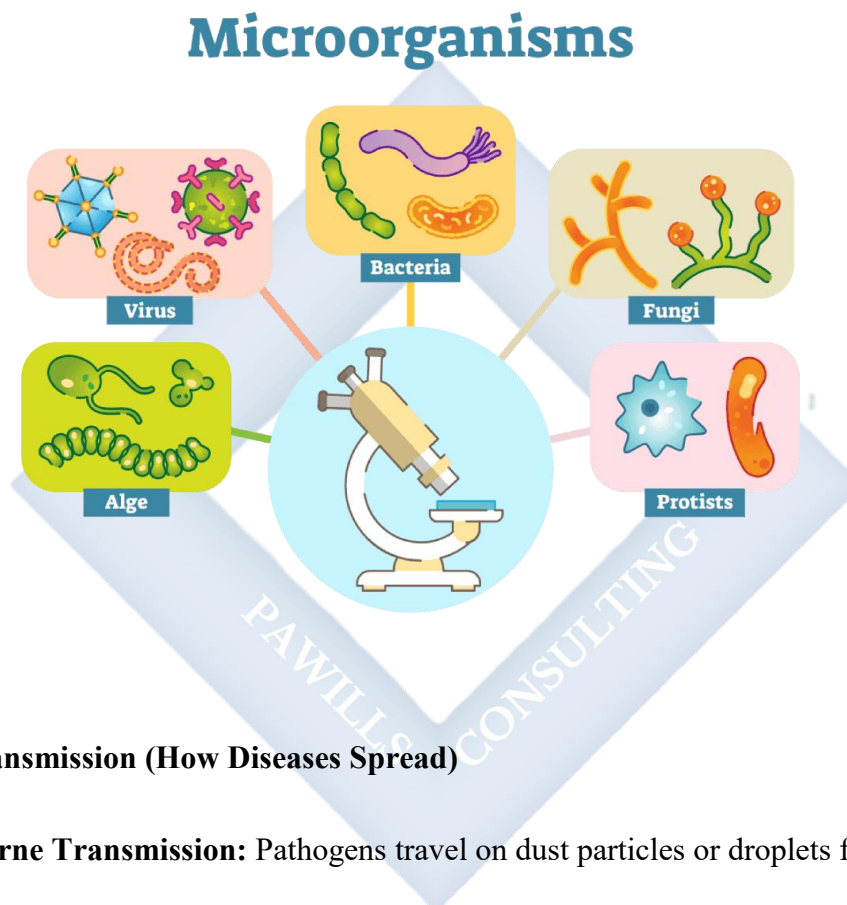
1. Define "microorganism" and "pathogen."
2. List the four (4) main types of pathogens (virus, bacteria, fungi, protozoa).
3. List three (3) main ways diseases are transmitted.
4. Explain "vaccination" as a method of prevention.
5. List five (5) personal hygiene practices to prevent disease.

Lesson Content:

1. Microorganisms and Pathogens

- **Microorganism (or Microbe):** A living organism that is too small to be seen with the naked eye. They are found everywhere.
- **Types:**
 1. **Bacteria:** Unicellular organisms. (Most are harmless, but some cause disease).
 2. **Viruses:** The *smallest* microbes. (All are parasitic and cause disease).
 3. **Fungi:** (e.g., Yeast, Mould).
 4. **Protozoa:** Unicellular "animal-like" microbes.

- **Pathogen:** A microorganism that *causes* disease.
 - *Virus* → causes COVID-19, Influenza (flu), HIV/AIDS, Measles.
 - *Bacteria* → causes Cholera, Typhoid, Tuberculosis.
 - *Fungi* → causes Ringworm, Athlete's Foot.
 - *Protozoa* → causes Malaria (Plasmodium), Dysentery (Amoeba).



2. Disease Transmission (How Diseases Spread)

1. **Air-borne Transmission:** Pathogens travel on dust particles or droplets from a cough or sneeze.
 - *Examples:* COVID-19, Tuberculosis, Common Cold.
2. **Water-borne Transmission:** Drinking water contaminated with sewage or feces.
 - *Examples:* Cholera, Typhoid.
3. **Vector-borne Transmission:** An animal (the "vector") carries the pathogen from one person to another.
 - *Vector:* Anopheles Mosquito → *Disease:* Malaria.
 - *Vector:* Rat → *Disease:* Lassa Fever.

4. **Food-borne Transmission:** Eating food that is contaminated or poorly cooked.
5. **Direct Contact / Blood-borne:** Touching an infected person, or contact with infected blood.
 - *Examples:* Ringworm (contact), HIV/AIDS (blood).

3. Prevention and Control

- **A. Vaccination (Immunization):**

- **Definition:** A **vaccine** is a *dead* or *weakened* version of a pathogen (or a piece of it) that is injected into the body.
- **How it works:** The body's **immune system** "sees" the weak pathogen and learns to build **antibodies** (soldiers) to fight it.
- The immune system then *remembers* this pathogen. If the *real, strong* pathogen enters the body later, the immune system is already prepared and destroys it immediately.
- **Result:** It provides **immunity** (protection) *without* causing the disease.
- *Examples:* Polio vaccine, Measles vaccine, COVID-19 vaccine.

- **B. Hygiene Practices (Personal and Environmental):**

1. **Hand Washing:** The *most important* hygiene practice. Wash hands with soap and water after using the toilet and before eating.
2. **Clean Water:** Boil water before drinking, or use water purification.
3. **Proper Waste Disposal:** (Week 8) Keep the environment clean to prevent rats and flies.
4. **Vector Control:** Sleep under an insecticide-treated mosquito net; clear gutters of stagnant water.
5. **Food Safety:** Wash fruits/vegetables; cook food (especially meat) thoroughly.
6. **Quarantine/Isolation:** Separating sick people to stop them from spreading the disease.

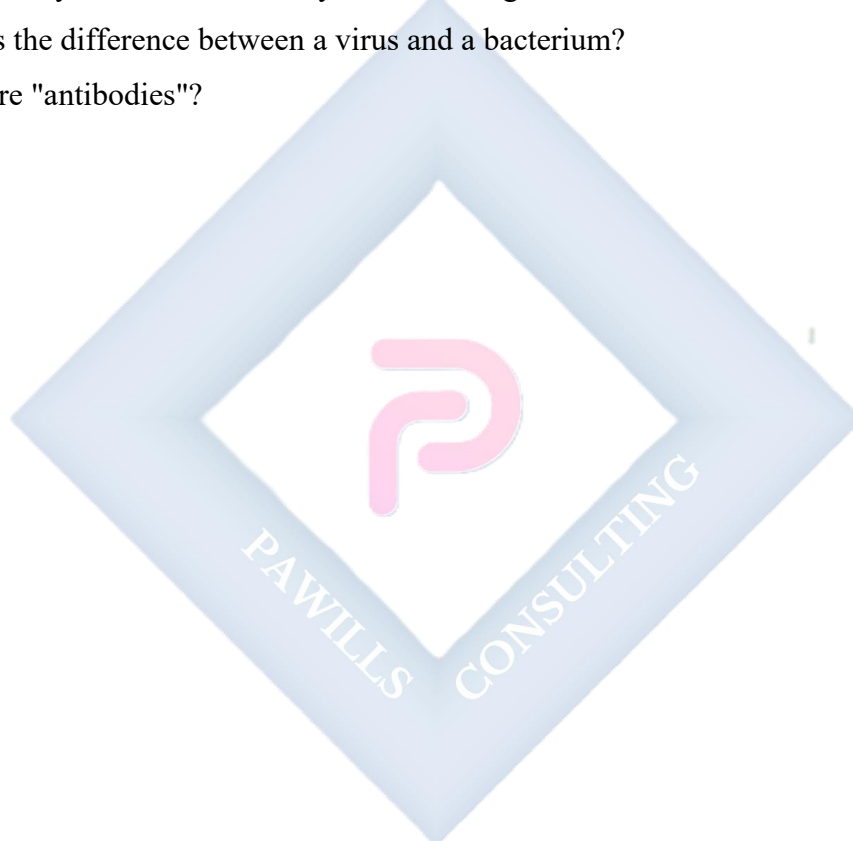
Evaluation:

1. What is a pathogen?
2. List the four (4) main types of microorganisms.
3. What is a "vector"? Give one example.

4. How does a vaccine work?
5. What is the *single most important* personal hygiene practice to prevent disease?

Assignment:

1. Define Pathogen, Vector, and Vaccination.
2. Create a table with three columns: Disease (Malaria, Cholera, Tuberculosis), Pathogen Type (Protozoa, Bacteria, etc.), and Method of Transmission (Vector, Water, etc.).
3. Explain, in your own words, why hand washing is so effective.
4. What is the difference between a virus and a bacterium?
5. What are "antibodies"?



WEEK 11: REVISION

Topic: Revision

Content: Review of all topics in Term 2.

Note: This week is dedicated to a comprehensive review of all topics covered in the Second Term to prepare for the examination.

Revision Checklist:

1. The Cell (Week 1):

- Cell Theory (3 points).
- Organelles & Functions (Nucleus, Mitochondrion, Cell Wall, Chloroplast, Cell Membrane, Vacuole).
- Plant vs. Animal Cell (3 differences).

2. Levels of Organization (Week 2):

- Cell → Tissue → Organ → Organ System → Organism.
- 4 Animal Tissues (Epithelial, Connective, Muscle, Nervous).
- Organ Systems (Digestive, Circulatory, Respiratory, etc.).

3. Heredity (Week 3):

- Heredity, Gene, Trait.
- Inherited (Blood group) vs. Acquired (Scar) traits.
- Variation (differences) and its importance.
- Dominant (T) vs. Recessive (t) genes.

4. Adaptation (Week 4):

- Habitat (home) and Adaptation (survival feature).
- Structural (body part - e.g., gills, camel's hump).
- Behavioral (action - e.g., migration, hibernation).

5. Chemistry (Weeks 7, 8):

- Atom Structure (Proton +, Neutron 0, Electron -).
- Element (1 type of atom), Compound (chemically bonded), Mixture (physically mixed).

- Symbols (H, O, C) and Formulae (H_2O , CO_2).
- Physical Change (reversible, no new substance - e.g., melting ice).
- Chemical Change (irreversible, new substance - e.g., burning wood).

6. Water and Air (Week 9):

- Water (H_2O), Air (N_2 78%, O_2 21%).
- Water Purification (Coagulation, Sedimentation, Filtration, Disinfection - Chlorine).
- Air Pressure (weight of air) - Crushing Can experiment.

7. Diseases (Week 10):

- Pathogen (causes disease - Virus, Bacteria, Fungi, Protozoa).
- Transmission (Air, Water, Vector).
- Vaccination (weak pathogen builds immunity).
- Hygiene (Hand washing).

Practical Revision:

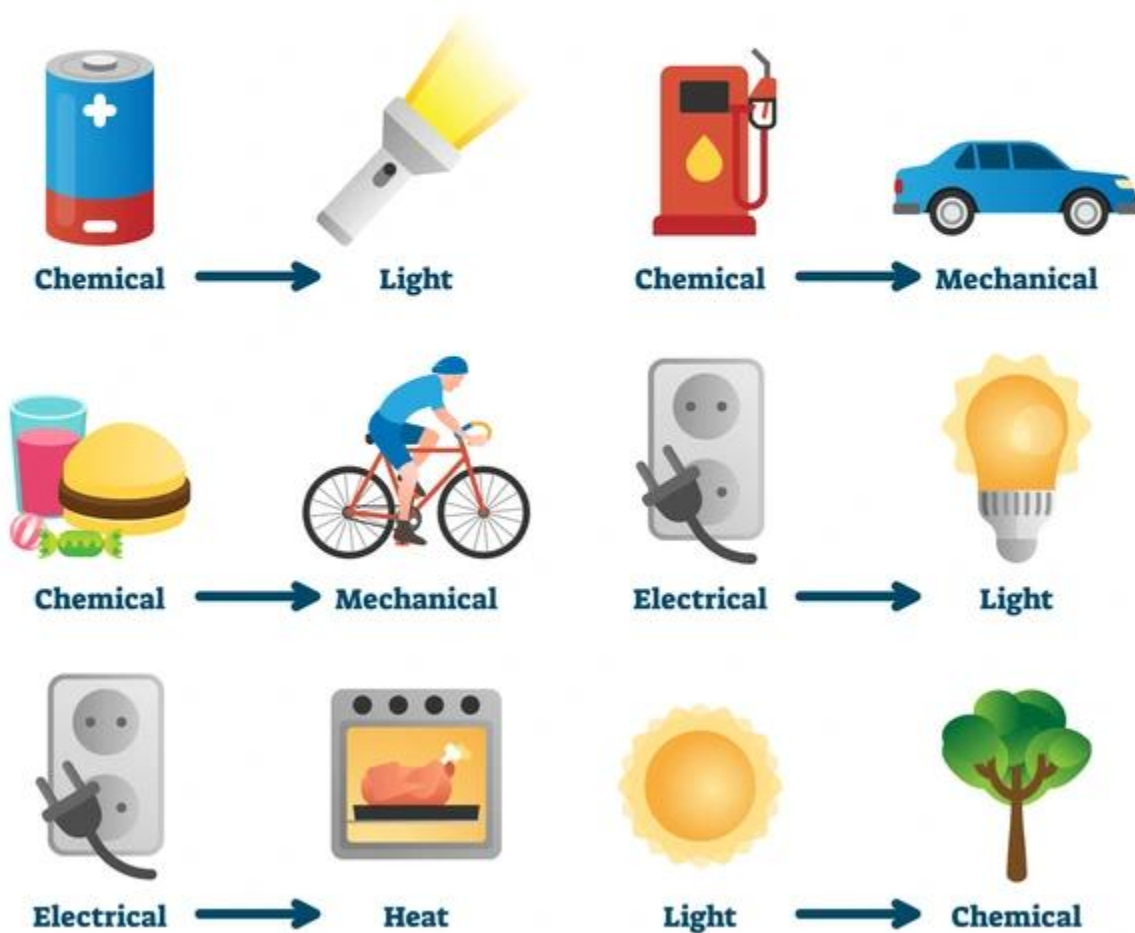
- Drawing and labeling plant/animal cells.
- Solving a simple dominant/recessive problem.
- Classifying changes as physical or chemical.
- Listing diseases and how they are transmitted.

THIRD TERM - JSS 3 INTERMEDIATE SCIENCE LESSON NOTES



WEEK 1: ENERGY TRANSFORMATION

ENERGY TRANSFORMATIONS



Content: (I) Conversion from one form to another (II) Energy conservation (III) Simple demonstrations.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Energy" and list at least five different forms (Heat, Light, Sound, Chemical, Electrical, Mechanical).
2. Define "Energy Transformation" (or Conversion).

3. State the **Principle of Conservation of Energy**.
4. Give at least three everyday examples of energy transformation, identifying the "input" and "output" forms.
5. Describe a simple demonstration that shows energy transformation.

Lesson Content:

1. Review: Forms and Sources of Energy

- **Energy:** Energy is the capacity or ability to do work. Work is done when a force causes an object to move. The standard unit of energy is the **Joule (J)**.
- **Forms of Energy (Recap):**
 - **Chemical Energy:** Stored in bonds of atoms and molecules (e.g., food, batteries, fuel).
 - **Electrical Energy:** The movement of electrons (e.g., electricity in wires).
 - **Mechanical Energy:** Energy of motion (Kinetic) or position (Potential).
 - **Heat (Thermal) Energy:** The vibration of atoms in a substance.
 - **Light (Radiant) Energy:** Energy from electromagnetic waves.
 - **Sound Energy:** Energy from vibrations that travel as waves.

2. Energy Transformation (Conversion)

- **Definition:** Energy Transformation is the process of **changing energy from one form into another**.
- We do not "use up" energy; we simply convert it into a form we need. In every conversion, some energy is also "lost" or wasted, usually as **heat**.

3. Examples of Energy Transformation: It is crucial to be able to trace the path of energy.

1. A Battery-Powered Torch:

- Chemical Energy (stored in battery) → Electrical Energy (in wires) → Light Energy (from the bulb) + Heat Energy (wasted).

2. A Human Running:

- Chemical Energy (stored in food) → Mechanical Energy (Kinetic energy of moving muscles) + Heat Energy (body gets warm).

3. **A Solar Panel System:**

- Light Energy (from the sun) → Electrical Energy (in the panel) → (e.g., powers a fan) → Mechanical Energy (blades spinning).

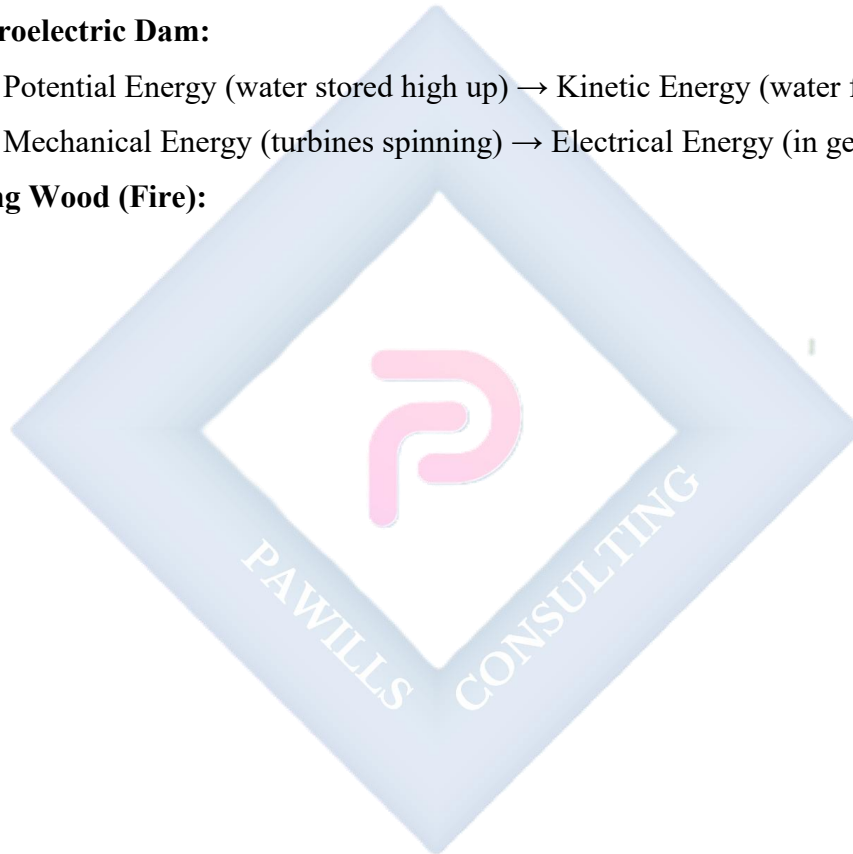
4. **An Electric Fan:**

- Electrical Energy (from socket) → Mechanical Energy (blades spinning) + Sound Energy + Heat Energy (motor gets warm).

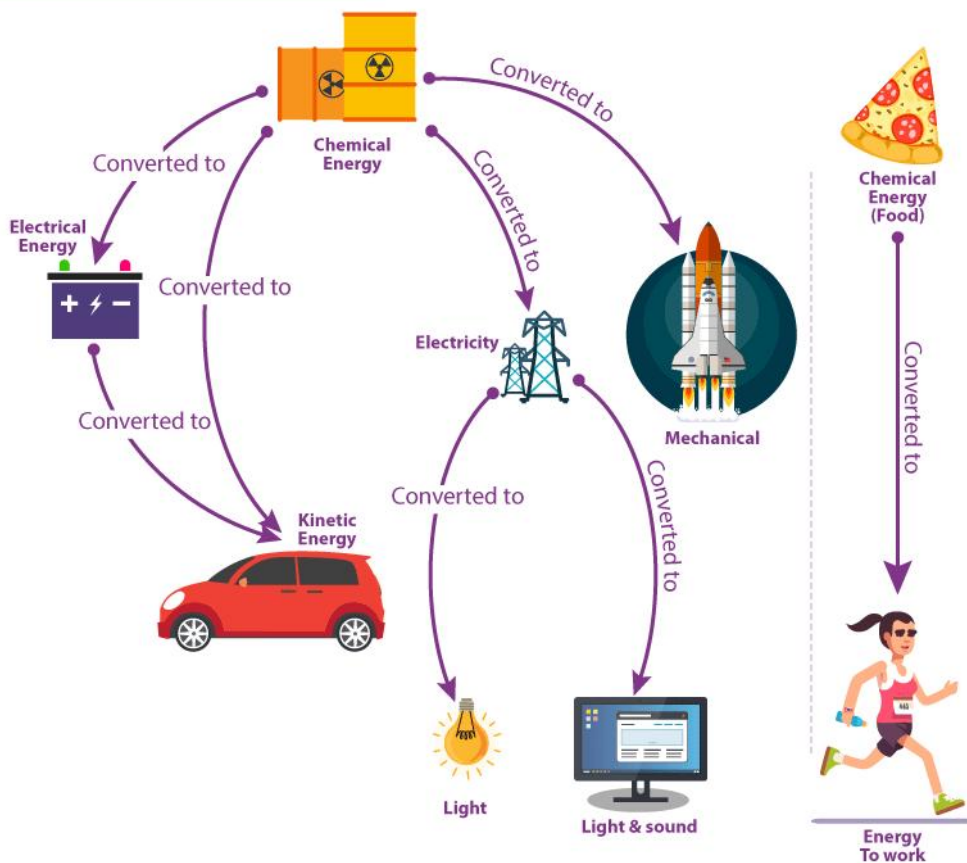
5. **A Hydroelectric Dam:**

- Potential Energy (water stored high up) → Kinetic Energy (water falling) → Mechanical Energy (turbines spinning) → Electrical Energy (in generators).

6. **Burning Wood (Fire):**



ENERGY CONVERSION



- Chemical Energy (stored in wood) → Heat Energy + Light Energy.

4. The Principle of Conservation of Energy

- This is a fundamental law of physics.
- **Statement:** "Energy **cannot** be created or destroyed, but it **can** be transformed (converted) from one form to another."
- **Meaning:** The total amount of energy in the universe is constant. When you use a torch, the chemical energy in the battery is not "gone"; it has simply *changed* into an equal total amount of light and heat energy.

5. Simple Demonstrations

- **A. The Dropped Ball (Potential to Kinetic):**

- Hold a ball high up. It has stored Potential Energy (due to its height).
- Drop the ball. As it falls, it loses height (loses Potential) and gains speed (gains Kinetic Energy).
- Just before it hits the ground, all its Potential Energy has transformed into Kinetic Energy.

- **B. A Simple Pendulum:**

-
- Pull a pendulum bob to one side (Position A). It has maximum Potential Energy.
- Release it. It swings down, and at the bottom (Position B), its energy is maximum Kinetic Energy.
- It swings up to the other side (Position C), converting its Kinetic Energy back into Potential Energy.
- This shows a continuous transformation between potential and kinetic energy.

Evaluation:

1. What is energy transformation?
2. State the Principle of Conservation of Energy.
3. Describe the energy transformation in an electric fan.
4. Describe the energy transformation when a person eats food and then claps their hands.
5. In a pendulum, where is the Potential Energy at its maximum? Where is the Kinetic Energy at its maximum?

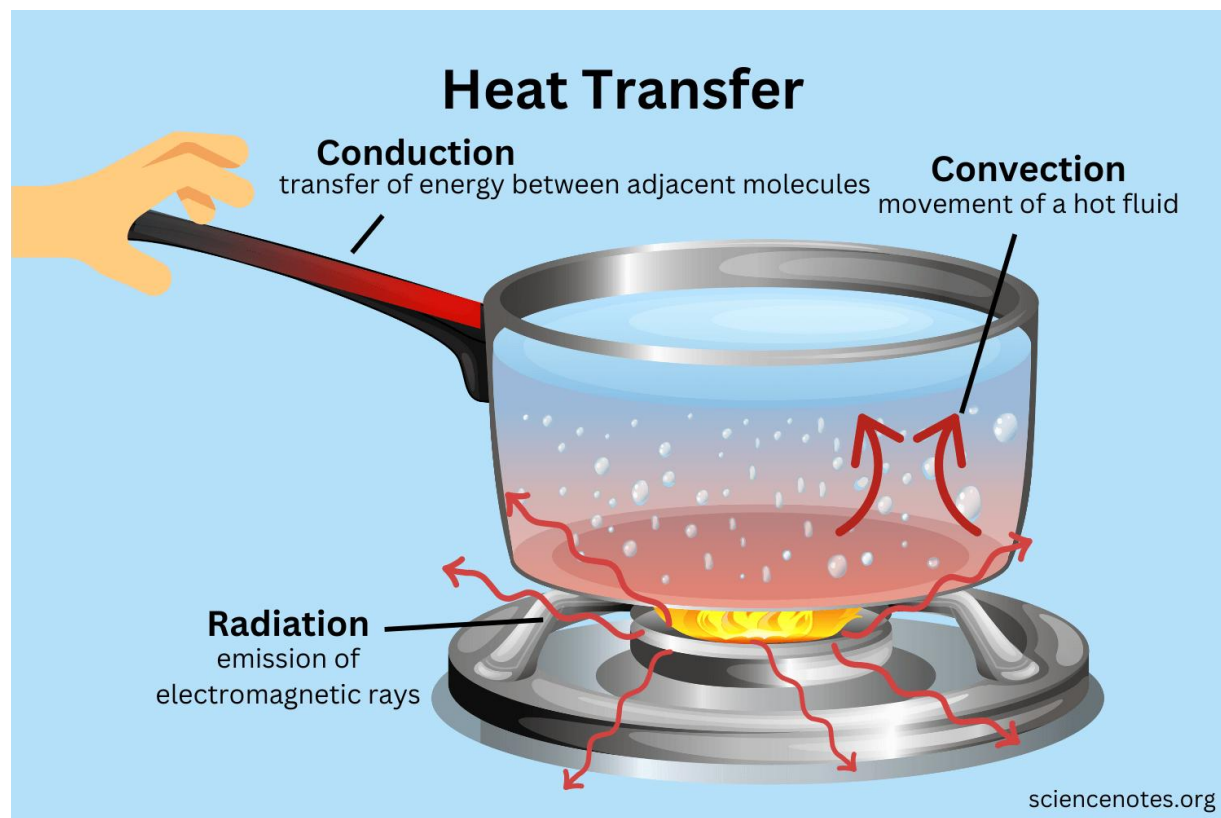
Assignment:

1. Define "Energy" and "Energy Transformation."
2. Write down the Principle of Conservation of Energy.
3. For each of the following, write the energy transformation path (e.g., Chemical → ...):
 - (a) A car engine burning petrol to move.
 - (b) A solar-powered calculator.
 - (c) A person shouting.

- (d) A windmill generating electricity.
4. Why does a bouncing ball not bounce back to the same height you dropped it from? (Hint: "wasted" energy).



WEEK 2: HEAT TRANSFER



Content: (I) Conduction, convection, radiation (II) Practical examples (III) Insulators and conductors.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Heat" as a form of energy.
2. Define and differentiate between the three methods of heat transfer: Conduction, Convection, and Radiation.
3. Give a clear example of each method from daily life.
4. Define "Conductor" and "Insulator" of heat, and give two examples of each.
5. Explain the principle behind a vacuum flask.

Lesson Content:

1. Definition of Heat

- **Heat (Thermal Energy):** Heat is a form of energy that is transferred from a *hotter* object to a *colder* object. Heat energy is the total kinetic energy of the atoms or molecules in a substance.
- Heat *always* flows from hot to cold.

2. Methods of Heat Transfer There are three ways heat can move:

• A. Conduction:

- **Definition:** The transfer of heat through a substance by **direct contact**, from molecule to molecule, without the molecules themselves moving from their position. They just vibrate and "pass" the energy along.
- **Medium:** Primarily occurs in **Solids**.
- **Example 1:** If you put a metal spoon in hot tea, the handle of the spoon will eventually become hot. The heat *conducted* from the tea, up the spoon, particle by particle.
- **Example 2:** Frying an egg. The heat from the stove conducts to the frying pan, which conducts to the egg.

• B. Convection:

- **Definition:** The transfer of heat through **fluids (liquids and gases)** by the *actual movement* of the fluid itself.
- **Process (Convection Current):**
 1. The fluid at the bottom (e.g., water in a pot) gets heated.
 2. When it gets hot, it **expands** and becomes **less dense** (lighter).
 3. This hot, light fluid *rises*.
 4. The *colder, denser* fluid at the top *sinks* to take its place.
 5. This cold fluid gets heated, and the cycle repeats. This circular movement is a **convection current**.
- **Examples:**
 1. **Boiling Water:** The water at the bottom rises, and the cooler water at the top sinks.

2. **Sea Breeze:** During the day, the land heats up faster than the sea. The hot air over the land rises, and the cool air from the sea blows in to replace it (a sea breeze).

- **C. Radiation:**

- **Definition:** The transfer of heat in the form of **electromagnetic waves** (specifically, infrared waves).
- **Medium:** Requires **NO MEDIUM** (no solids, liquids, or gases). It can travel through a vacuum (empty space).
- **Examples:**
 1. **The Sun:** The heat from the sun travels millions of kilometers through the vacuum of space to warm the Earth.
 2. **A Fire:** You can feel the heat from a bonfire from a distance, even if the air between you is not moving. The heat radiates to your skin.
 3. A microwave oven.

3. Conductors and Insulators

- **Conductors (Heat):** Materials that allow heat to pass through them *easily* and *quickly*.
 - *Examples:* All **metals** (like silver, copper, aluminum, iron). This is why cooking pots are made of metal.
- **Insulators (Heat):** Materials that do **not** allow heat to pass through them easily. They are poor conductors.
 - *Examples:* **Wood, Plastic, Glass, Rubber, Fabric (cloth), Air** (trapped air is a very good insulator). This is why pot handles are made of plastic or wood.
- **Application: The Vacuum (Thermos) Flask**
 - A vacuum flask (or "thermos") is designed to stop all three forms of heat transfer.
 - It has two walls (like a bottle inside a bottle).
 - **Conduction:** The stopper (lid) is made of plastic (an insulator). The walls are separated, so heat can't conduct.
 - **Convection:** The space between the two walls is a **vacuum** (all the air has been removed). Since there is no fluid (air), convection cannot occur.

- **Radiation:** The two walls are coated with a shiny, silver layer (like a mirror) to



Types of Heat Transfer

Conduction



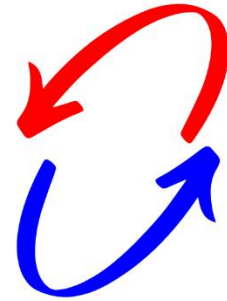
Touching a hot pot

Radiation



The Sun's heat rays

Convection



Convection currents under the Earth's surface

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reflect heat radiation (keeping hot things hot and cold things cold).

Evaluation:

1. Define Conduction, Convection, and Radiation.
2. Why does heat transfer mainly occur through conduction in solids and convection in liquids?
3. What is a heat insulator? Give two examples.
4. How does heat from the sun reach the Earth? (Radiation).
5. Explain how a convection current works when boiling water.

Assignment:

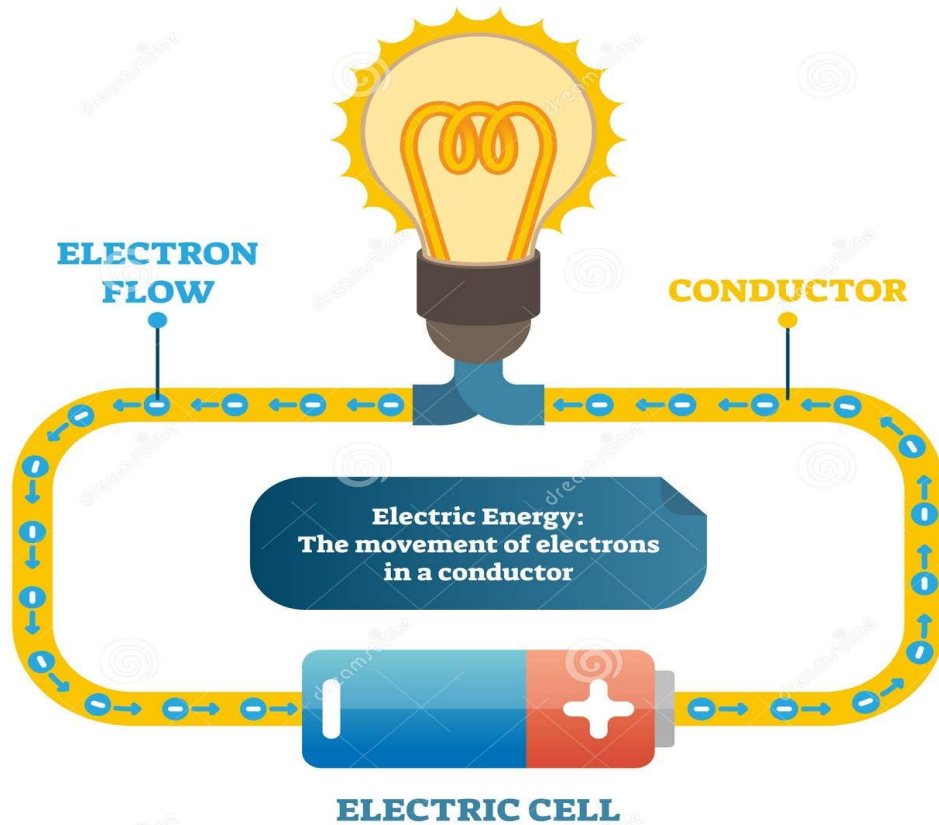
1. Create a table with three columns: Conduction, Convection, Radiation. For each, write its definition, the medium it travels through (solid, fluid, vacuum), and one example.

2. Why are the handles of cooking pots made of plastic or wood? (Use the word "insulator").
3. Why do we wear thick, "fluffy" clothes in cold weather? (Hint: Fluffy clothes trap *air*, which is an insulator).
4. Explain the two ways a vacuum flask stops heat loss by conduction and convection.



WEEK 3: ELECTRICAL ENERGY

⚡ ELECTRIC ENERGY ⚡



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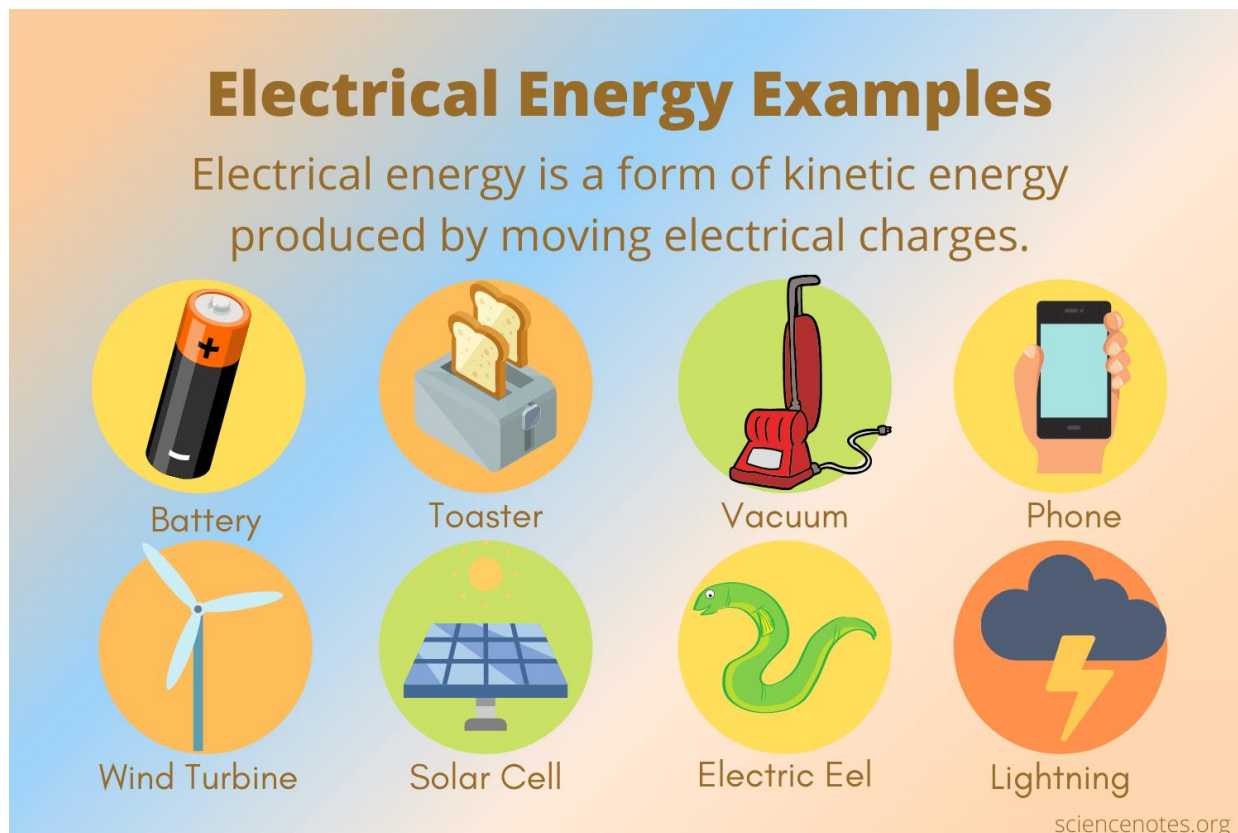
Content: (I) Meaning and sources (II) Electric circuit and symbols (III) Domestic wiring and safety (IV) Conservation of electricity.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "electrical energy" and "electric current."
2. List three (3) sources of electrical energy.
3. Define "electric circuit" and draw a simple circuit using standard symbols (cell, bulb, switch, wire).
4. Differentiate between parallel and series circuits.

5. List five (5) electrical safety rules for the home.
6. List three (3) ways to conserve electricity.

Lesson Content:



1. Meaning and Sources of Electrical Energy

- **Electrical Energy:** The energy caused by the *movement of electric charges* (usually electrons).
- **Electric Current (I):** The *flow* of these electric charges through a material (a conductor). It is measured in **Amperes (A)**.
- **Sources of Electrical Energy (How we "make" electrons move):**
 1. **Chemical Energy (Cells/Batteries):** Chemical reactions inside a battery (e.g., in your remote, phone, or car) push electrons through a wire.

2. **Generators (Mechanical Energy):** This is how large-scale power is made. A turbine (moved by water, steam from burning gas, or wind) spins a large magnet near coils of wire, forcing electrons to move.
3. **Solar Energy (Light Energy):** Solar panels (photovoltaic cells) convert light energy directly into electrical energy.
4. **Other sources:** Wind (wind turbines), Hydro (dams), Nuclear.

2. Electric Circuit and Symbols

- **Electric Circuit:** A complete, closed **path** that allows electric current to flow. If the path is broken (e.g., a cut wire or an open switch), it is an **open circuit**, and the current stops.
- **Components and Symbols:**
 - **Connecting Wire:** ———
 - **Cell (Battery - DC source):** —|— (long line is positive, short line is negative)
 - **Bulb (Lamp):** —(X)— or —(O)—
 - **Switch (Open):** — / — (path is broken)
 - **Switch (Closed):** — . — (path is complete)
- **Simple Circuit Diagram:**
- **Types of Circuits:**
 - **Series Circuit:** Components are connected in a *single path*, one after another.
 - *Advantage:* Simple to build.
 - *Disadvantage:* If *one* bulb breaks, the entire circuit is open, and all other bulbs go out. (Like old Christmas lights).
 - **Parallel Circuit:** Components are connected on *separate branches* or paths.
 - *Advantage:* If *one* bulb breaks, the other branches are still complete, so the other bulbs *stay lit*.
 - This is how homes are wired.

3. Domestic Wiring (Home Wiring) and Safety

- Your house is wired in **parallel**.
- There are three main wires:

1. **Live (L):** The "hot" wire that carries the high voltage. (Brown colour).
2. **Neutral (N):** The wire that completes the circuit, carrying current back. (Blue colour).
3. **Earth (E):** The safety wire. It connects the metal case of an appliance to the ground. If the live wire touches the case, the current flows safely to the earth, blowing a fuse and preventing an electric shock. (Green/Yellow stripes).

- **Electrical Safety Rules:**

1. **NEVER** touch switches, sockets, or appliances with wet hands. (Water conducts electricity).
2. **DO NOT** overload a socket by plugging in too many high-power appliances. (Can cause a fire).
3. **DO NOT** use appliances with frayed or broken wires.
4. **NEVER** poke metal objects (like forks or fingers) into a socket.
5. **ALWAYS** switch off an appliance before unplugging it.
6. **DO NOT** fly kites near overhead power lines.

4. Conservation of Electricity Conserving (saving) electricity saves money and protects the environment (by burning less fuel).

1. **Turn off** lights and fans when you leave a room.
2. Use **energy-efficient bulbs** (LEDs) instead of old incandescent bulbs.
3. Unplug electronics (like TVs and chargers) when not in use.
4. Ensure your refrigerator door is sealed tightly.

Evaluation:

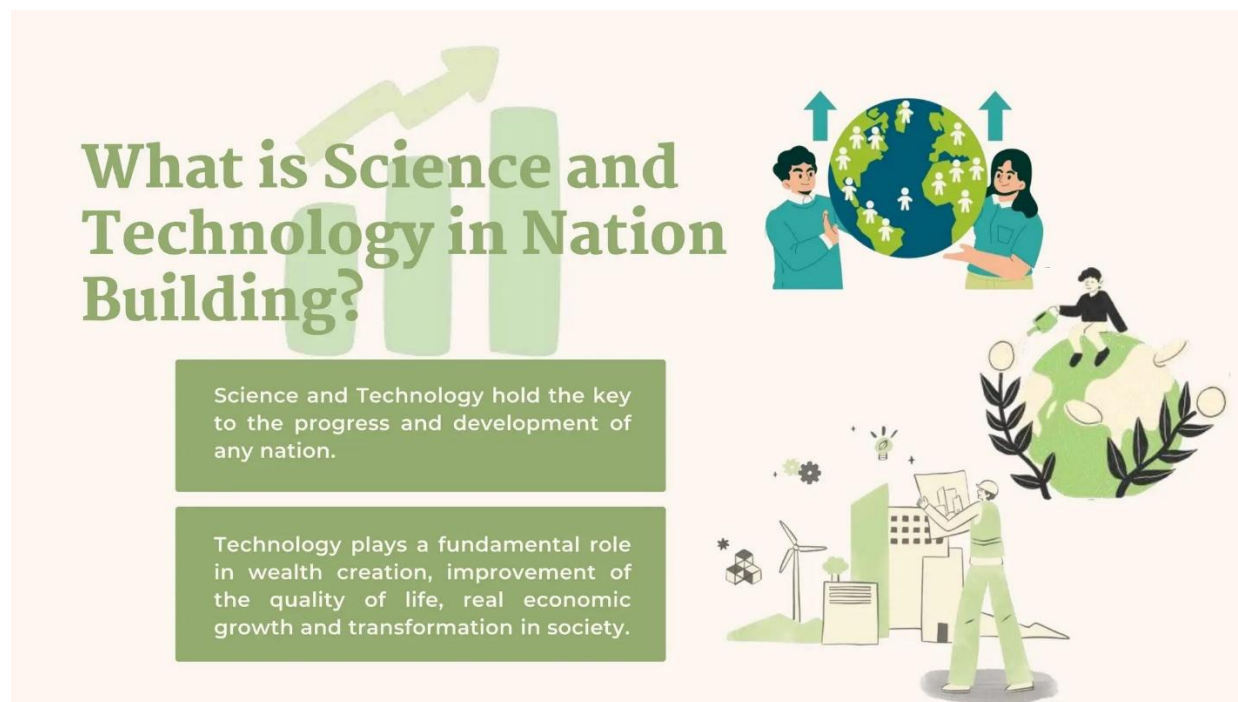
1. What is electric current?
2. Draw the circuit symbols for a cell, a bulb, and an open switch.
3. Why are houses wired in parallel, not in series?
4. What is the purpose of the Earth wire in a plug?
5. List three (3) important electrical safety rules.

Assignment:

1. Define electrical energy and list three (3) sources.
2. Draw a simple circuit diagram showing a battery (2 cells), a closed switch, and two bulbs connected in **series**.
3. Draw a simple circuit diagram showing a battery (2 cells), a closed switch, and two bulbs connected in **parallel**.
4. Why should you never touch a switch with wet hands?
5. List three (3) ways your family can conserve electricity at home.



WEEK 4: SCIENCE, TECHNOLOGY, AND NATIONAL DEVELOPMENT



Content: (I) Relationship among science, technology, and development (II) Role of research (III) Inventions and innovators.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Science" and "Technology."
2. Explain the relationship between Science and Technology.
3. Define "National Development."
4. Explain how science and technology contribute to national development (e.g., in medicine, agriculture).
5. Differentiate between an "invention" and a "discovery."
6. Name at least one local and one international innovator/inventor.

Lesson Content:

1. Defining Key Terms

- **Science:** A systematic process of acquiring **knowledge** about the natural world through **observation, hypothesis, and experimentation**. Science is about *knowing why* something happens. (e.g., *Science* discovers the laws of aerodynamics).
- **Technology:** The **application** of scientific knowledge for *practical purposes*. It is the creation of **tools, machines, and processes** to solve human problems. Technology is about *knowing how* to make something happen. (e.g., *Technology* uses aerodynamics to build an aeroplane).
- **National Development:** The overall improvement in the quality of life and standard of living of the citizens of a country. This includes growth in the economy (GDP), healthcare, education, infrastructure (roads, electricity), and security.

2. Relationship Among Science, Technology, and Development

This is a "feedback loop":

1. **Science** (Knowledge) → leads to new ideas.
 2. **Technology** (Application) → uses those ideas to create tools and solutions.
 3. **Development** (Growth) → these tools improve society (e.g., better medicine, faster communication).
 4. This development (e.g., a richer country) can then *fund* more **Science** (e.g., build universities and labs), leading to new discoveries, and the cycle continues.
- *Example:*
 - Science: Biologists discover that a certain mould (Penicillium) kills bacteria.
 - Technology: Engineers and chemists use this knowledge to mass-produce the drug **Penicillin**.
 - Development: Penicillin is used to cure infections, saving millions of lives and improving national health.

3. Role of Research and Development (R&D)

- **Research** (the "S" in R&D) is the engine of science.
- **Development** (the "D" in R&D) is the engine of technology.

- A country that invests heavily in R&D (in universities, research institutes) will discover new solutions and create new products, leading to faster national development.
- Countries like the USA, China, and South Korea are leaders in technology because they spend billions on R&D.

4. Contributions to National Development

- **Agriculture:** Technology (tractors, fertilizers, irrigation) increases food production, leading to food security.
- **Medicine:** Science (understanding diseases) and Technology (vaccines, MRI scanners, new drugs) lead to longer, healthier lives.
- **Communication:** Technology (satellites, fibre optics, mobile phones) makes business faster and connects people.
- **Industry:** Technology (robotics, automation) makes manufacturing more efficient.

5. Inventions and Innovators

- **Discovery:** Finding something that *already exists* in nature. (e.g., Newton *discovering* gravity).
- **Invention:** Creating something *new* that did not exist before. (e.g., Thomas Edison *inventing* the lightbulb).
- **Innovation:** Taking an existing invention and *improving* it significantly or finding a new use for it.
- **Notable Innovators/Inventors:**
 - *International:*
 - **Thomas Edison:** Lightbulb, phonograph.
 - **Alexander Graham Bell:** Telephone.
 - **Tim Berners-Lee:** Invented the World Wide Web (Internet).
 - *Local (Nigerian):*
 - **Prof. Wole Soboyejo:** Scientist known for work in materials science.
 - **Jelani Aliyu:** Designed the Chevrolet Volt (a revolutionary electric car).

- *Many local innovators* who create machines for garri processing, palm oil extraction, etc., solving local problems.

Evaluation:

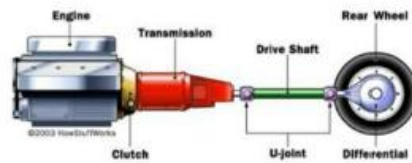
1. What is the main difference between science and technology?
2. Define "National Development."
3. Explain the relationship between science, technology, and development.
4. What is the difference between an "invention" and a "discovery"?
5. Name one international inventor and one Nigerian innovator.

Assignment:

1. Define Science, Technology, and National Development in your own words.
2. Using an example (like the mobile phone), explain the link between science, technology, and development.
3. Why is "Research and Development" (R&D) important for a country?
4. Choose one sector (Agriculture, Medicine, or Communication). Explain two ways technology has contributed to its development in Nigeria.
5. Who is Jelani Aliyu, and what is he famous for?

WEEK 6: MACHINES AND MECHANISMS

MACHINES AND MECHANISMS



Content: (I) Simple and compound machines (II) Functions and uses (III) Mechanical advantage (IV) Practical demonstration.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Machine" and "Work" (in the scientific sense).
2. Define "Effort," "Load," and "Pivot."
3. List the six (6) types of simple machines.
4. Differentiate between a simple and a compound machine.
5. Define "Mechanical Advantage" (MA) and state its formula.

Lesson Content:

1. Definition of Work and Machine

- **Work (Scientific Definition):** Work is done when a **force (Effort)** moves an object (the **Load**) over a **distance**.
- **Machine:** A machine is any device that makes work *easier* by:
 1. Reducing the amount of **Effort** needed (e.g., a lever).
 2. Changing the **direction** of the force (e.g., a pulley).
 3. Increasing the **speed** or **distance** (e.g., a bicycle gear).

2. Key Terms

- **Load (L):** The force or weight that you are trying to move (the *resistance*).
- **Effort (E):** The force *you* apply to the machine to move the load.
- **Pivot (or Fulcrum):** The fixed point around which a lever turns.

3. Simple Machines vs. Compound Machines

- **Simple Machine:** A device that has one or two parts and makes work easier. There are **six** types.
- **Compound Machine:** A machine made up of *two or more simple machines* working together.
 - *Examples:* A **wheelbarrow** (a lever + a wheel & axle), a **bicycle** (levers, pulleys, wheels & axles), a **car**.

4. The Six (6) Types of Simple Machines

1. **Lever:** A rigid bar that turns around a fixed point (the pivot).
 - *Examples:* See-saw, crowbar, bottle opener, wheelbarrow, scissors (two levers).
2. **Pulley:** A wheel with a groove that a rope or chain passes over. Used for lifting.
 - *Examples:* A flagpole, a crane, a well.
3. **Wheel and Axle:** A large wheel fixed to a smaller rod (the axle). When one turns, the other turns.
 - *Examples:* A doorknob, a car's steering wheel, a tap handle.
4. **Inclined Plane:** A flat, sloping surface. It allows you to move an object to a greater height with less effort (but over a longer distance).

- *Example:* A ramp, a slide.
- 5. **Wedge:** Two inclined planes put together, used to split things apart.
 - *Examples:* An axe, a knife, a nail, your front teeth.
- 6. **Screw:** An inclined plane wrapped around a cylinder (a spiral). Used to hold things together or lift.
 - *Examples:* A screw, a bolt, a car jack.

5. Mechanical Advantage (MA)

- **Definition:** This is the *advantage* you get from using a machine. It is a number that tells you how many times the machine *multiplies* your effort.
- **Formula:**
 - Mechanical Advantage (MA) = Load / Effort
- **Example:**
 - I want to lift a **Load** of 100 Newtons.
 - Using a lever, I only have to push down with an **Effort** of 20 Newtons.
 - My MA = 100 N / 20 N = 5.
 - This means the machine multiplied my effort by 5, making the work 5 times easier.

6. Practical Demonstration (Lever)

- Use a ruler (lever) and a pencil (pivot).
- Place a heavy book (load) on one end (e.g., at 5cm).
- Place the pencil (pivot) under the 10cm mark.
- Push down on the other end (effort) to lift the book.
- Now, move the pivot *closer* to the load (e.g., at 8cm). It becomes *easier* to lift (MA has increased).

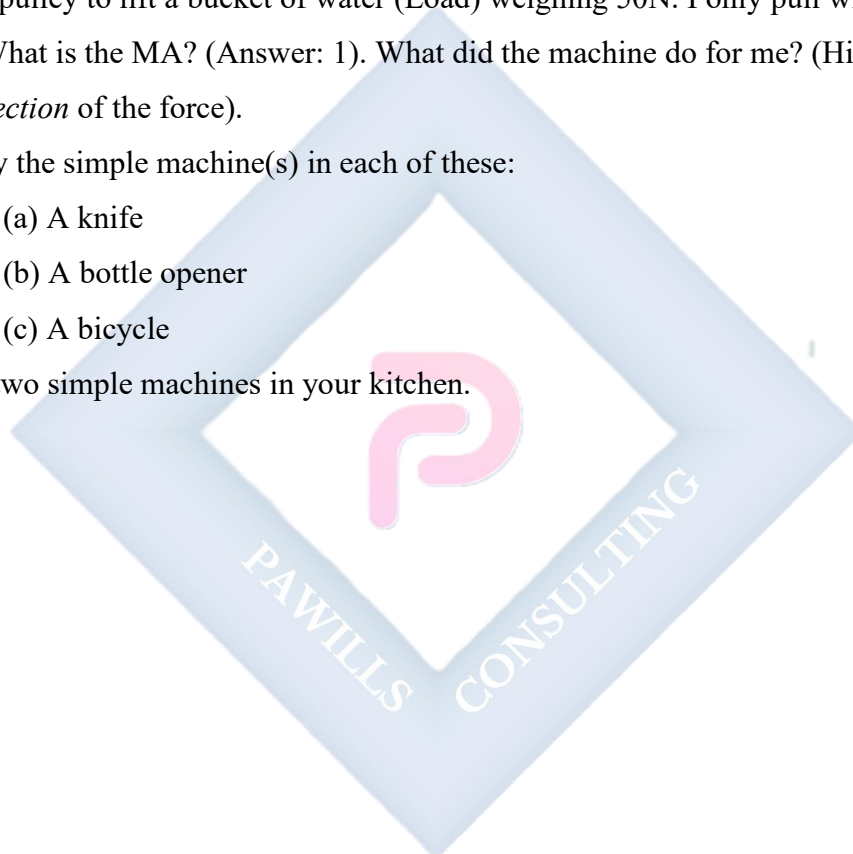
Evaluation:

1. What is a machine?
2. What is the scientific definition of work?

3. List the six (6) types of simple machines.
4. What is the difference between a simple and a compound machine?
5. What is the formula for Mechanical Advantage (MA)?

Assignment:

1. Define Load, Effort, and Pivot.
2. What is Mechanical Advantage?
3. I use a pulley to lift a bucket of water (Load) weighing 50N. I only pull with an Effort of 50N. What is the MA? (Answer: 1). What did the machine do for me? (Hint: it changed the *direction* of the force).
4. Identify the simple machine(s) in each of these:
 - (a) A knife
 - (b) A bottle opener
 - (c) A bicycle
5. Name two simple machines in your kitchen.



WEEK 8: ICT AND SCIENTIFIC ADVANCEMENT



Content: (I) Role of ICT in science (II) Data collection and storage (III) Modern communication tools (IV) Internet and global learning.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "ICT."
2. Explain four (4) major roles of ICT in modern science.
3. Describe how ICT is used for data collection and storage.
4. Explain the importance of the internet for global scientific collaboration.

Lesson Content:

1. What is ICT?

- **ICT** stands for **Information and Communication Technology**.
- It refers to all technologies used to **handle information** (store, process, retrieve) and **communicate** (share).
- *Examples:* Computers, smartphones, the Internet, software (like Microsoft Excel), email, satellites.

2. Role of ICT in Science ICT has *revolutionized* science, making it faster, more accurate, and more collaborative.

1. **Data Collection and Storage:**

- *Before ICT:* Scientists wrote observations in a notebook (slow, small).
- *With ICT:* Digital sensors (like thermometers, weather stations, satellite scanners) collect *massive* amounts of data (called "Big Data") every second, 24/7.
- This data is stored on powerful computers, hard drives, or **cloud storage**, where it is safe and can be accessed from anywhere.

2. **Data Analysis and Simulation:**

- *Before ICT:* Scientists did calculations by hand (slow, prone to errors).
- *With ICT:* Powerful computers and software (like Excel) can analyze millions of data points in seconds to find patterns.
- **Simulation:** Scientists can create *computer models* to test things that are too big, too small, too dangerous, or too slow. (e.g., simulate a car crash, model climate change over 100 years, or see how a virus might spread).

3. **Communication and Collaboration:**

- *Before ICT:* Scientists shared ideas by writing letters (took months).
- *With ICT:* A scientist in Nigeria can instantly share data with a scientist in Japan via **email**, **video conferencing** (Zoom), or online research portals. This speeds up discovery.

4. **Access to Information (Global Learning):**

- *Before ICT*: A student had to use only the textbooks in their local library.
- *With ICT*: The **Internet** gives a student access to almost all of human knowledge—articles, videos, lectures, and research papers from around the world—instantly.

3. The Internet and Global Learning

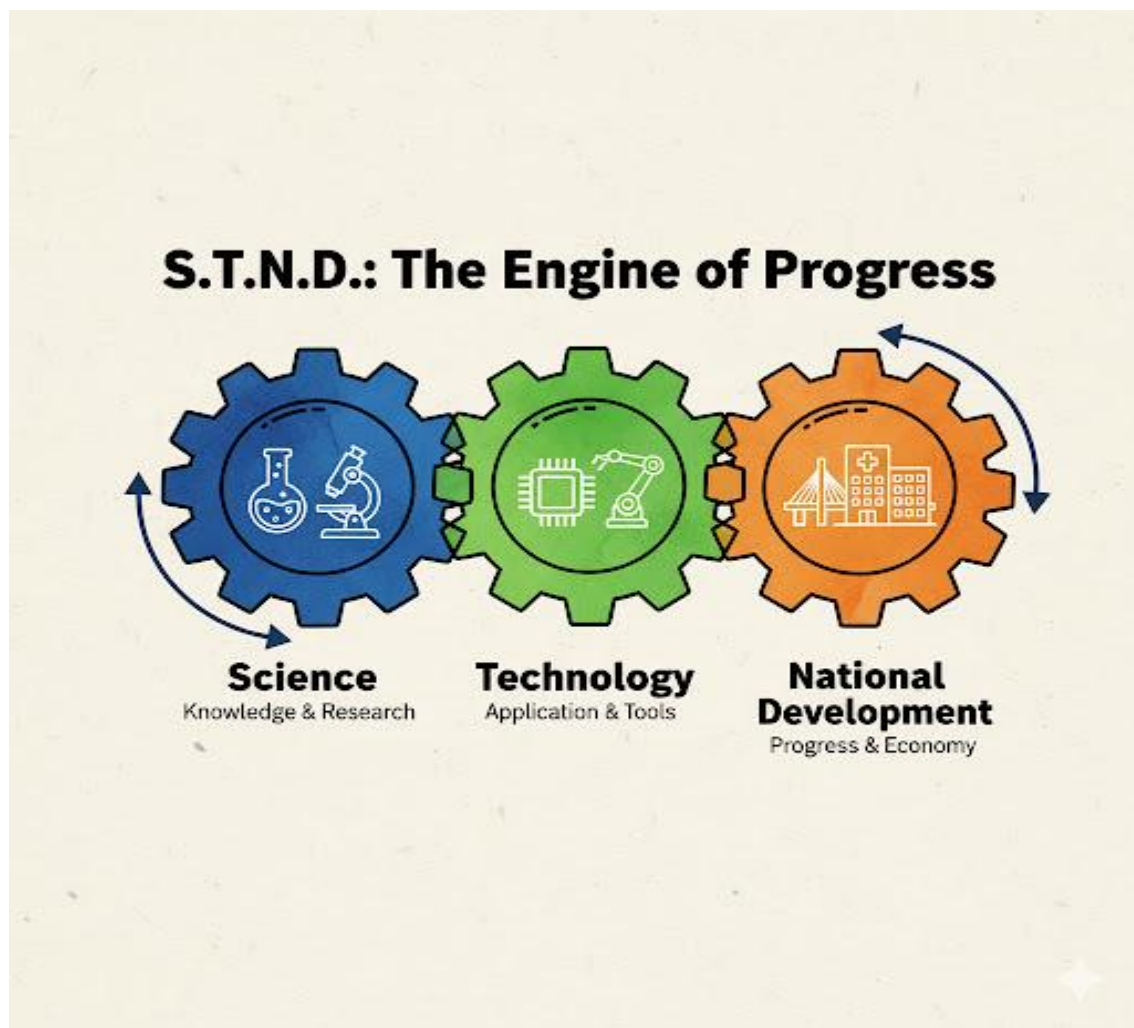
- The Internet is a global network of computers. It is the *most powerful* tool in ICT for science.
- It allows for:
 - **Global Collaboration**: Scientists on different continents can work on the same project (e.g., the Human Genome Project).
 - **Open Access**: Many research papers are published online for free.
 - **Crowdsourcing**: Scientists can ask the public to help, e.g., by reporting bird sightings or running small simulations on their home computers.

Evaluation:

1. What does ICT stand for?
2. List three (3) roles of ICT in scientific advancement.
3. What is a "simulation"?
4. How has ICT changed data *storage*?
5. What is the most important ICT tool for global collaboration? (The Internet).

Assignment:

1. Define ICT in your own words.
2. Explain how a scientist can use ICT for "Data Collection."
3. Explain how a scientist can use ICT for "Data Analysis."
4. Imagine you are a scientist in Nigeria who just discovered a new plant that might cure malaria. How would you use ICT (email, internet) to share your findings with the world?
5. What is "cloud storage"?



WEEK 9: CAREER OPPORTUNITIES IN SCIENCE AND TECHNOLOGY

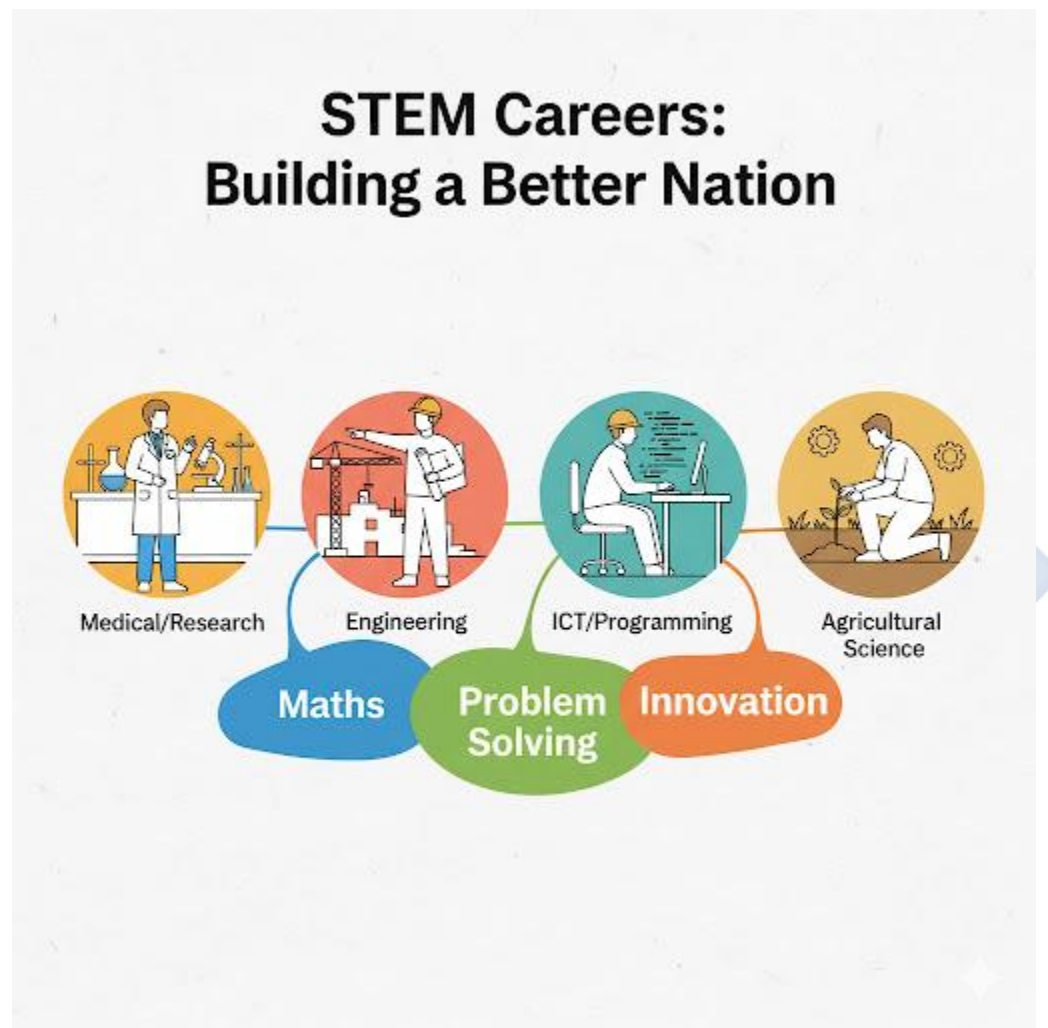
Content: (I) Careers in medical, engineering, ICT, and research fields (II) Skills required (III) Importance to national growth.

Performance Objectives: By the end of this lesson, students should be able to:

1. List at least two (2) careers in each of the following fields: Medical, Engineering, ICT, and Research.
2. Describe the main job function of at least three (3) of these careers.

3. List three (3) general skills required for a successful career in science and technology.
4. Explain why having skilled professionals in these fields is important for Nigeria's national growth.

Lesson Content:



1. What are Science and Technology Careers? These are jobs that require the knowledge of science, mathematics, and technology to solve problems and create new things. They are some of the most important jobs for building a modern country.

2. Career Fields and Examples

- **A. Medical Field (Health Sciences):**

- **Doctor (Physician):** Diagnoses and treats illnesses and injuries.
- **Pharmacist:** Prepares and dispenses medicines; an expert on drugs.
- **Medical Laboratory Scientist:** Runs tests on blood, tissue, and fluids to help doctors diagnose diseases.
- **Nurse:** Provides care for patients in hospitals and clinics.
- **Radiographer:** Operates X-ray, CT, and MRI scanners.
- **B. Engineering Field:**
 - **Civil Engineer:** Designs and builds infrastructure (roads, bridges, dams, large buildings).
 - **Electrical Engineer:** Designs electrical systems (power grids, electronics, computers).
 - **Mechanical Engineer:** Designs and builds machines (engines, generators, cars, robots).
 - **Chemical Engineer:** Designs processes to turn raw materials into useful products (e.g., refining petroleum, making soap, processing food).
- **C. Information & Comm. Technology (ICT) Field:**
 - **Software Developer (Programmer):** Writes the code that creates computer programs, apps, and websites.
 - **Network Administrator:** Manages and secures a company's or school's computer network.
 - **Data Scientist:** Analyzes large amounts of data to find patterns and make predictions.
 - **Cybersecurity Expert:** Protects computer systems from hackers and viruses.
- **D. Research Field (Pure Science):**
 - **Biologist/Biotechnologist:** Studies living things; may develop new medicines or crops.
 - **Chemist:** Studies substances; may invent new materials (like plastics or batteries).
 - **Physicist:** Studies energy and matter; may work in space exploration or energy.
 - **Geologist:** Studies the Earth; explores for oil, gas, and minerals.

3. Skills Required for S&T Careers

- **1. Strong Foundation Subjects:** You must be good at **Mathematics**, Physics, Chemistry, and Biology.
- **2. Problem-Solving:** The main job is to find solutions to difficult problems.
- **3. Analytical Skills:** The ability to look at data and understand what it means.
- **4. Creativity and Curiosity:** Thinking of new ideas and asking "Why?"
- **5. ICT Literacy:** Almost all science and technology jobs require strong computer skills.

4. Importance to National Growth

- A nation *cannot* develop without its own scientists, doctors, and engineers.
- **Health:** More doctors and nurses mean a healthier population, which is more productive.
- **Infrastructure:** Civil engineers build the roads and bridges needed to transport goods.
- **Economy:** ICT professionals and engineers create new companies (like Paystack, Flutterwave), create jobs, and solve local problems.
- **Self-Reliance:** When Nigeria has its own experts, it doesn't have to depend on other countries to build its infrastructure or solve its problems.

Evaluation:

1. What does a Civil Engineer do?
2. What does a Software Developer do?
3. List two careers in the medical field.
4. What is the most important school subject for almost all S&T careers? (Mathematics).
5. Explain one reason why these careers are important for Nigeria's development.

Assignment:

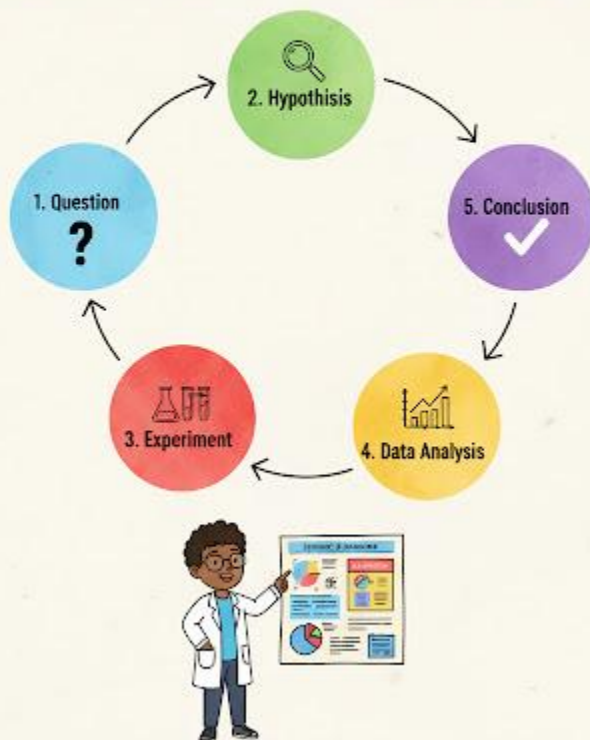
1. Choose *one* career from the four fields discussed (e.g., Pharmacist, Mechanical Engineer, etc.).
2. Write one paragraph describing what that person does.
3. List the JSS subjects you think are most important for that career.

4. Why did you choose this career? (What interests you about it?)
5. What is the difference between a Pharmacist and a Medical Laboratory Scientist?



WEEK 10: SCIENCE PROJECTS AND EXHIBITIONS

The Scientific Method: Your Path to Discovery.



Content: (I) Planning and execution (II) Presentation and teamwork (III) Assessment and reflection.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "science project" and state its purpose (to answer a question).
2. List the basic steps of the Scientific Method.
3. Explain the importance of teamwork in a science project.
4. Demonstrate the ability to plan a simple project (e.g., "What type of soil holds the most water?").
5. Explain how to present a project at an exhibition.

Lesson Content:

1. What is a Science Project?

- A science project is a practical, hands-on investigation where you use the **Scientific Method** to answer a question or solve a problem.
- It is not just building a model; it is an *experiment* designed to find an answer.

2. The Scientific Method (Steps for a Project):

1. **Ask a Question (Problem):** Start with something you are curious about.
 - *Bad Question:* "What is a plant?" (This is just a fact).
 - *Good Question:* "Does the *colour* of light affect how fast a plant grows?"
2. **Do Background Research:** Read books or use the internet to see what is already known.
3. **Form a Hypothesis:** Make an "educated guess" to answer your question.
 - *Hypothesis:* "I *predict* that the plant under the blue light will grow tallest."
4. **Design and Conduct an Experiment:** Plan a *fair test* to check your hypothesis.
 - *Fair Test:* You must have a **control group** (plant with normal light) and **variables**. You only change *one* thing (the **independent variable** - light colour). Everything else (water, soil, plant type) must be *kept the same* (the **constants**).
5. **Collect and Record Data:** Observe and *measure* what happens. Write it down in a table or chart.
6. **Analyze Data:** Look at your results. Did the blue light plant grow tallest?
7. **Draw a Conclusion:** State your final answer.
 - "My hypothesis was correct/incorrect. The plant under blue light grew tallest."
8. **Present Your Findings:** (See below).

3. Planning, Execution, and Teamwork

- **Planning:** This is the most important step. You must write down your question, hypothesis, and a list of all materials and steps.
- **Teamwork:** Most projects are too big for one person.

- **Roles:** Divide the work. (e.g., One person does research, one person gathers materials, one person takes measurements).
- **Collaboration:** Share ideas and listen to each other to find the best solution.

4. Presentation and Exhibition

- A science exhibition is where you *present* your project to others.
- **Presentation:** You must communicate your findings clearly. This usually involves:
 1. **A Display Board:** A large cardboard (like a triptych) that shows all the steps of your project (Title, Hypothesis, Method, Data, Conclusion).
 2. **The Experiment:** Your project materials, so you can demonstrate.
 3. **An Oral Presentation:** A 2-5 minute speech where you *explain* what you did and what you found.

5. Assessment and Reflection

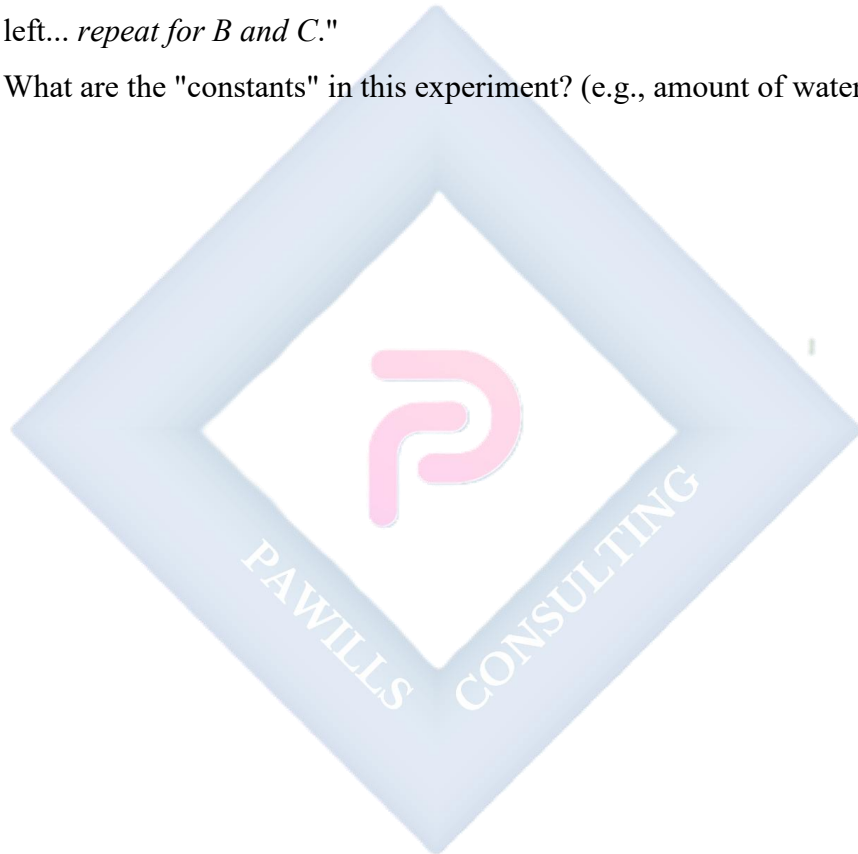
- Your project will be assessed (marked) on:
 - Creativity of the question.
 - Correct use of the Scientific Method.
 - Clarity of your data and results.
 - How well you present and explain your project.
- **Reflection:** Ask yourself: "What went wrong?" "What went right?" "What would I do differently next time?"

Evaluation:

1. What is a science project?
2. What is the "Scientific Method"?
3. What is a "hypothesis"?
4. What is a "fair test"?
5. Why is teamwork important?

Assignment:

1. List the seven (7) main steps of the Scientific Method in order.
2. What is the difference between an "independent variable" (what you change) and a "constant" (what you keep the same)?
3. **Plan a Simple Project:**
 - **Question:** "What brand of paper towel is the most absorbent?"
 - **Hypothesis:** "I predict that [Brand A] will be the most absorbent."
 - **How to Test (Method):** "I will (1) Get 3 brands, (2) Pour 20ml of water, (3) Place one sheet of [Brand A] on the water for 10 seconds, (4) Measure the water left... *repeat for B and C.*"
 - What are the "constants" in this experiment? (e.g., amount of water, time).



WEEK 11: REVISION

Topic: Revision

Content: Review of all topics in Term 3.

Note: This week is dedicated to a comprehensive review of all topics covered in the Third Term to prepare for the final JSS3 examination.

Revision Checklist:

1. Energy (Week 1):

- Principle of Conservation of Energy (cannot be created/destroyed).
- Energy Transformation (e.g., Battery: Chemical → Electrical → Light + Heat).
- Potential (stored) vs. Kinetic (moving) energy.

2. Heat Transfer (Week 2):

- **Conduction:** Solids, direct contact (hot spoon).
- **Convection:** Fluids (liquids/gases), movement of fluid (boiling water, sea breeze).
- **Radiation:** Waves, no medium (Sun, fire).
- Conductors (metals) vs. Insulators (wood, plastic, air).

3. Electrical Energy (Week 3):

- Circuit (path), Symbols (cell, bulb, switch).
- Series (one path) vs. Parallel (many paths - used in homes).
- Safety (wet hands, no overloading, Earth wire).
- Conservation (turn off lights, use LED).

4. Science & Development (Week 4):

- Science (knowing) vs. Technology (doing/application).
- Link: Science → Technology → National Development.
- Invention (new) vs. Discovery (existing).

5. Machines (Week 7):

- Machine (makes work easier).
- 6 Simple Machines (Lever, Pulley, Wheel/Axle, Inclined Plane, Wedge, Screw).
- Compound Machine (2 or more simple machines).

- Mechanical Advantage (MA) = Load / Effort.

6. ICT & Science (Week 8):

- ICT (Info & Comm. Tech).
- Role in Science: Data collection, analysis (simulations), storage (cloud), communication (internet).

7. Careers in S&T (Week 9):

- Medical (Doctor, Pharmacist), Engineering (Civil, Electrical), ICT (Programmer), Research (Biologist).
- Skills: Maths, Problem-Solving.

8. Science Projects (Week 10):

- Scientific Method (Question, Hypothesis, Experiment, Data, Conclusion).
- Fair Test (change one variable, keep constants).

Practical Revision:

- Tracing energy transformations in 5 different devices.
- Identifying the 6 simple machines in a diagram of a bicycle.
- Designing a simple experiment to answer a question.